

CSE 6740: Computational Data Analysis

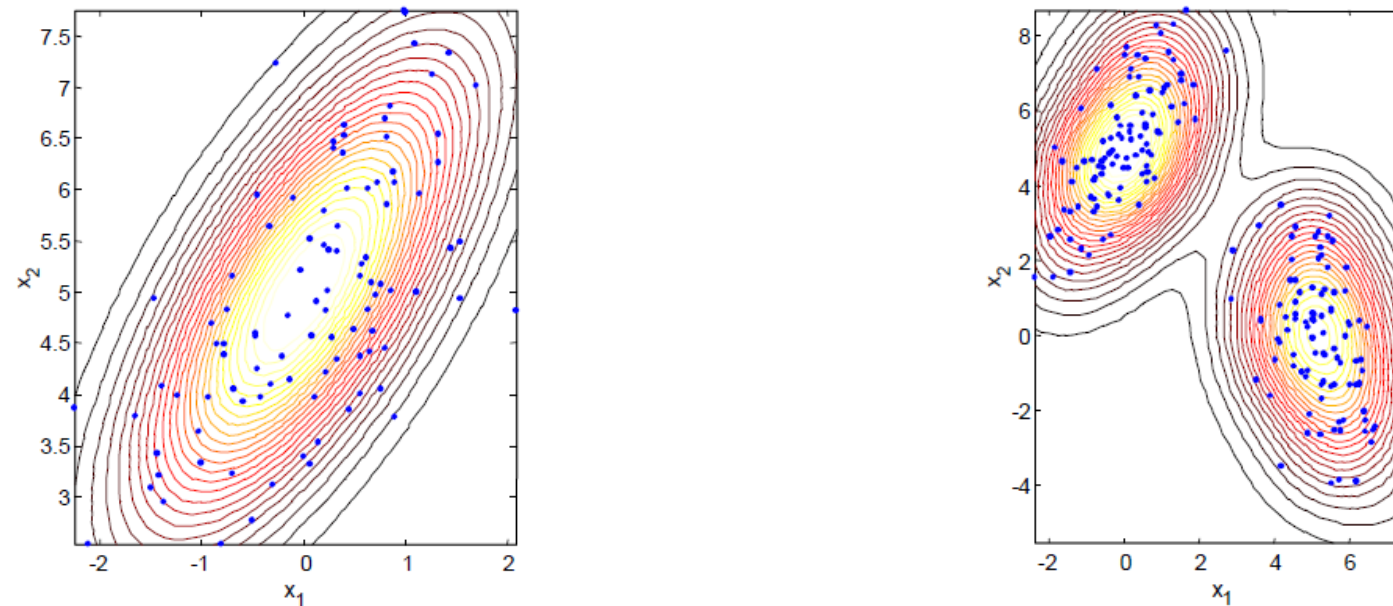
Spring 2026

Density Estimation

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02/05

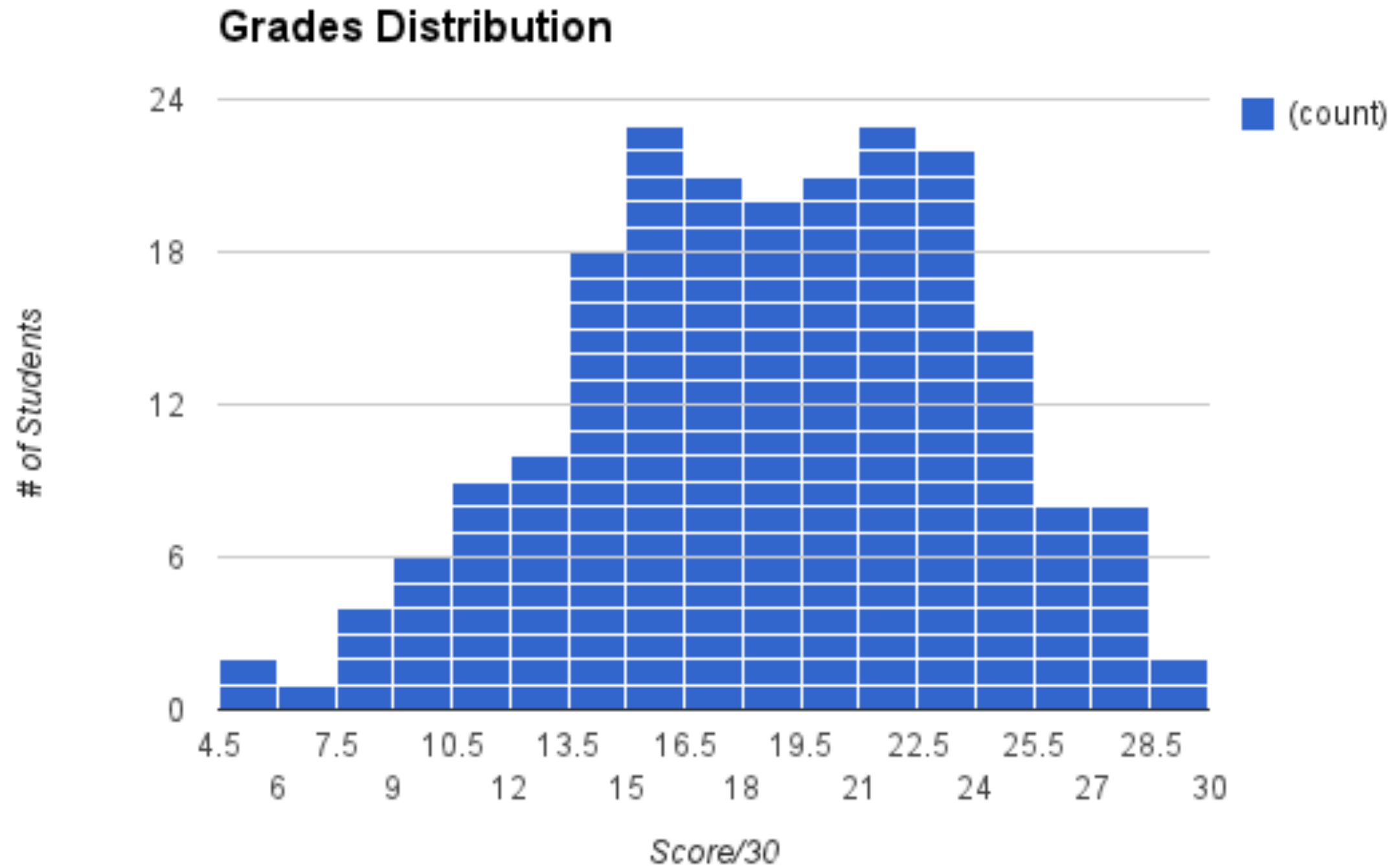
Why do we need density estimation?

- Learn more about the “shape” of the data cloud

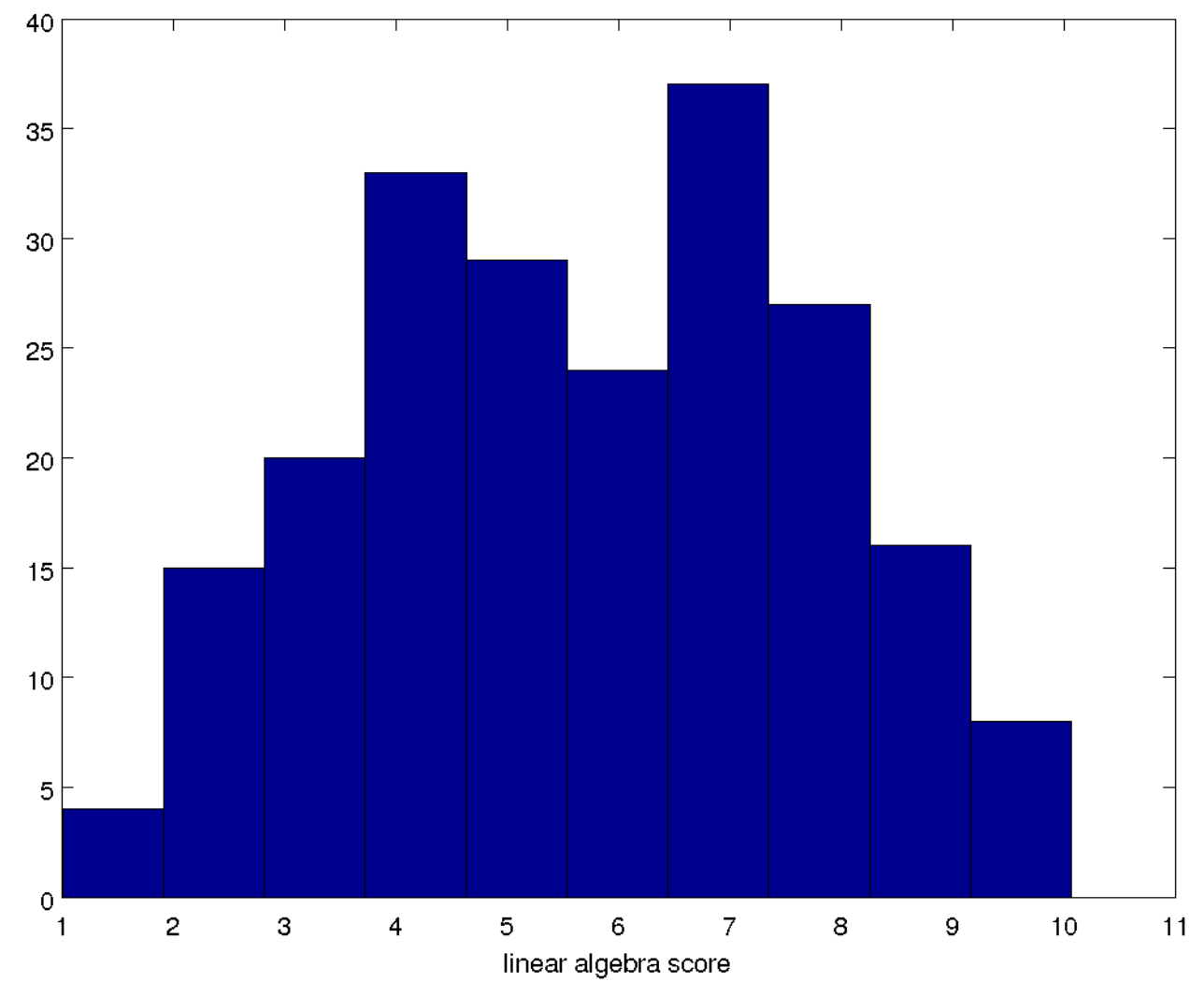
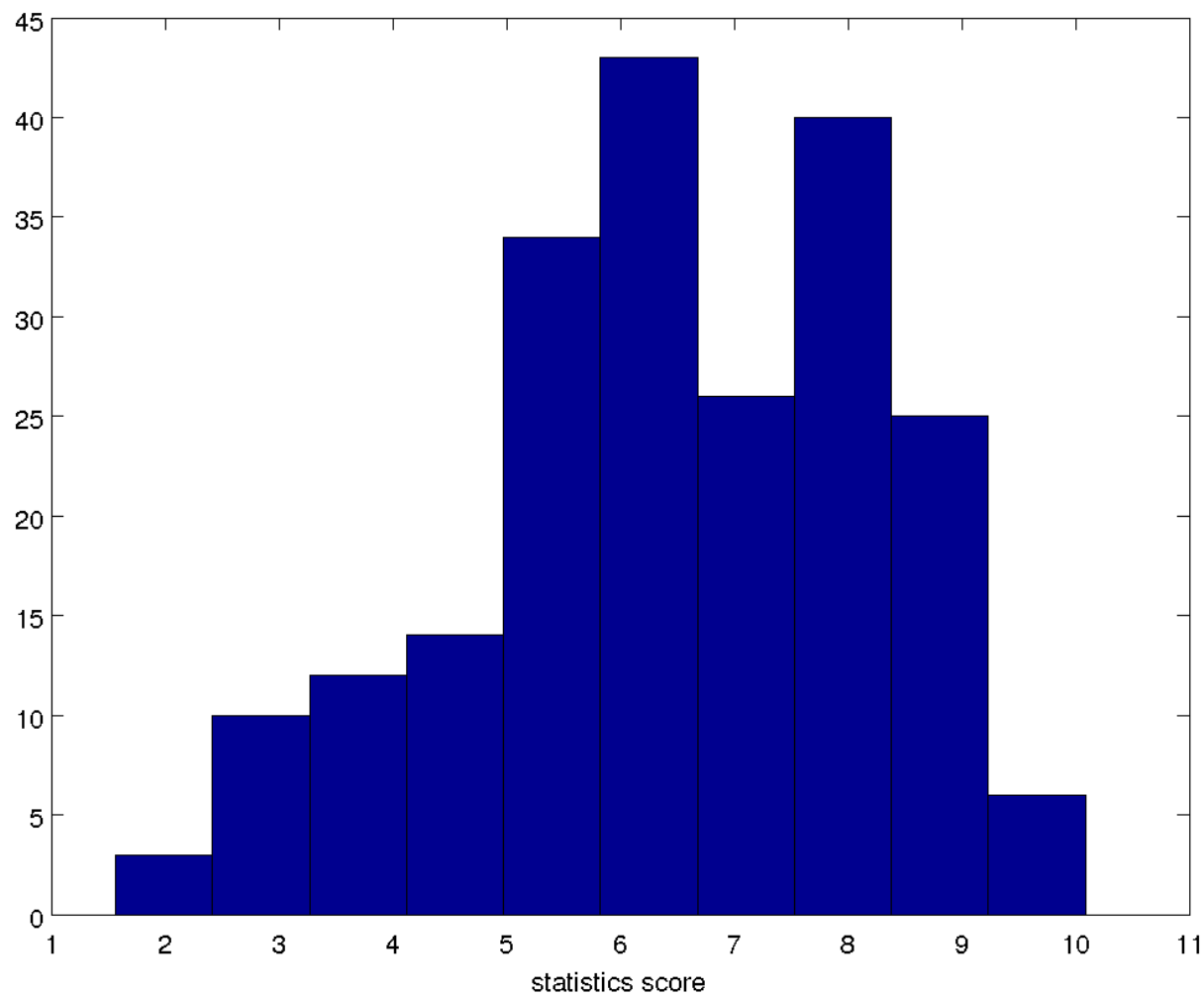


- Assess the likelihood of seeing a particular data point
 - Is this a typical data point? (high density value)
 - Is this an abnormal data point / outlier? (low density value)
- Building block for more sophisticated learning algorithms
 - Classification, regression, graphical models ...
 - A simple recommendation system

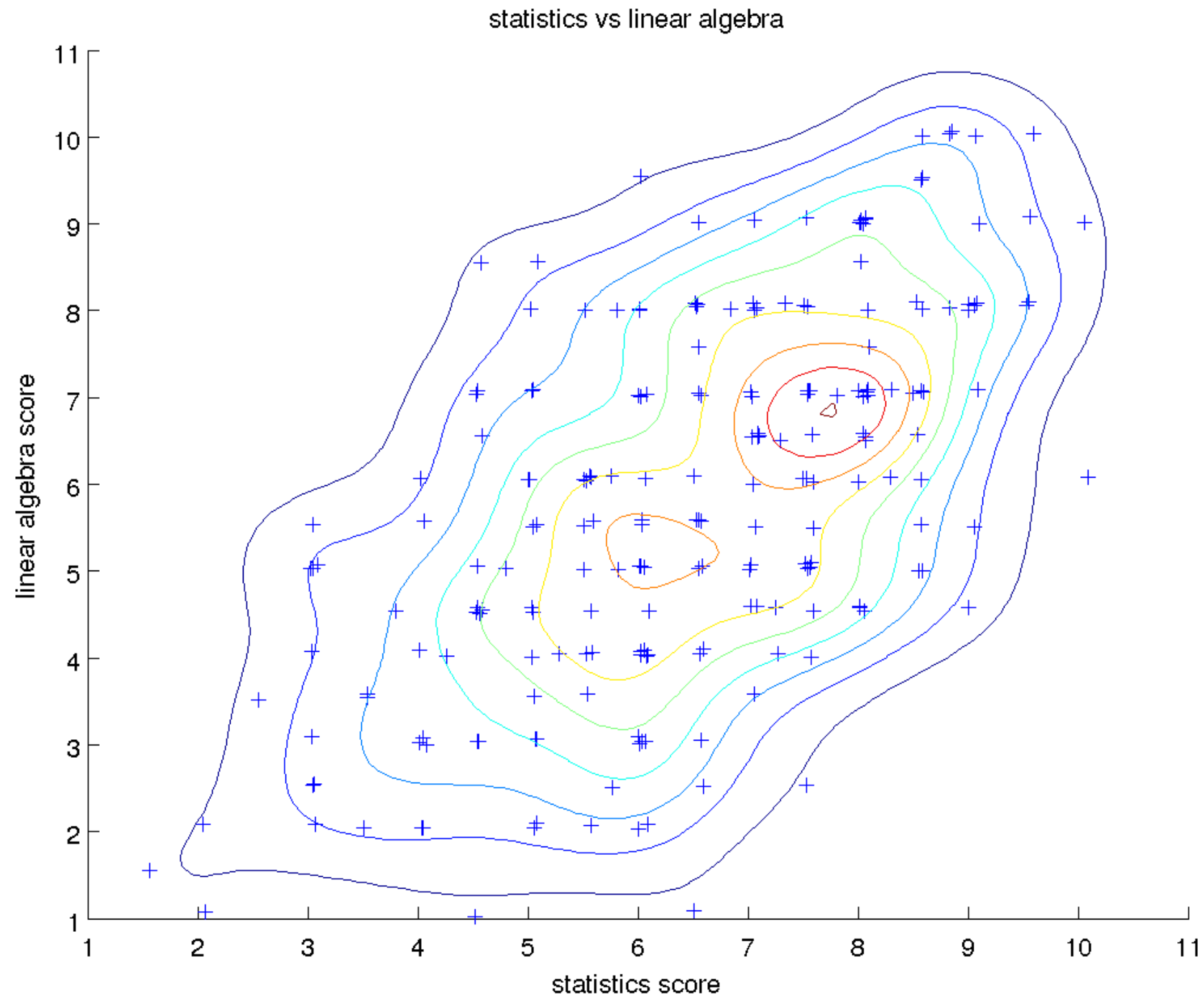
Example: test scores



Example: test scores (cont.)



Example: test scores (cont.)



Parametric models

- Models which can be described by a fixed number of parameters

- Discrete case: eg. Bernoulli distribution

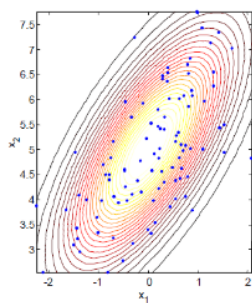
$$P(x|\theta) = \theta^x (1 - \theta)^{1-x}$$

one parameter, $\theta \in [0,1]$, which generate a family of models, $\mathcal{F} = \{P(x|\theta) \mid \theta \in [0,1]\}$,



- Continuous case: eg. Gaussian distribution in R^n

$$p(x|\mu, \Sigma) = \frac{1}{|\Sigma|^{\frac{1}{2}} (2\pi)^{\frac{n}{2}}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$



Two sets of parameters $\{\mu, \Sigma\}$, which again generate a family of models, $\mathcal{F} = \{p(x|\mu, \Sigma) \mid \mu \in R^n, \Sigma \in R^{n \times n} \text{ and PSD} \}$,

Estimation of parametric models

- A very popular estimator is the **maximum likelihood estimator (MLE)**, which is simple and has good statistical properties
- Assume that m data points $\mathcal{D} = \{x^1, x^2, \dots, x^m\}$ drawn **independently and identically (iid)** from some distribution $P^*(x)$
- Want to fit the data with a model $P(x|\theta)$ with parameter θ

$$\theta = \operatorname{argmax}_{\theta} \log P(\mathcal{D}|\theta) = \operatorname{argmax}_{\theta} \log \prod_{i=1}^m P(x^i|\theta)$$

Example problem

- Estimate the probability θ of landing in heads using a biased coin
- Given a sequence of m independently and identically distributed (iid) flips
 - Eg., $\mathcal{D} = \{x^1, x^2, \dots, x^m\} = \{1, 0, 1, \dots, 0\}, x^i \in \{0, 1\}$
- Model: $P(x|\theta) = \theta^x(1 - \theta)^{1-x}$
 - $P(x|\theta) = \begin{cases} 1 - \theta, & \text{for } x = 0 \\ \theta, & \text{for } x = 1 \end{cases}$
- Likelihood of a single observation x^i ?
 - $P(x^i|\theta) = \theta^{x^i}(1 - \theta)^{1-x^i}$



MLE for Biased Coin

- Objective function, log likelihood

$$\begin{aligned} l(\theta; \mathcal{D}) &= \log P(\mathcal{D}|\theta) = \log \theta^{n_h} (1 - \theta)^{n_t} \\ &= n_h \log \theta + (m - n_h) \log(1 - \theta) \end{aligned}$$

n_h : number of heads, n_t : number of tails

- Maximize $l(\theta; \mathcal{D})$ w.r.t. θ
- Take derivatives w.r.t. θ

$$\frac{\partial l}{\partial \theta} = \frac{n_h}{\theta} - \frac{(m - n_h)}{1 - \theta} = 0$$

$$\Rightarrow \hat{\theta}_{MLE} = \frac{n_h}{m} \text{ or } \hat{\theta}_{MLE} = \frac{1}{m} \sum_i x^i$$

Estimating Gaussian distribution

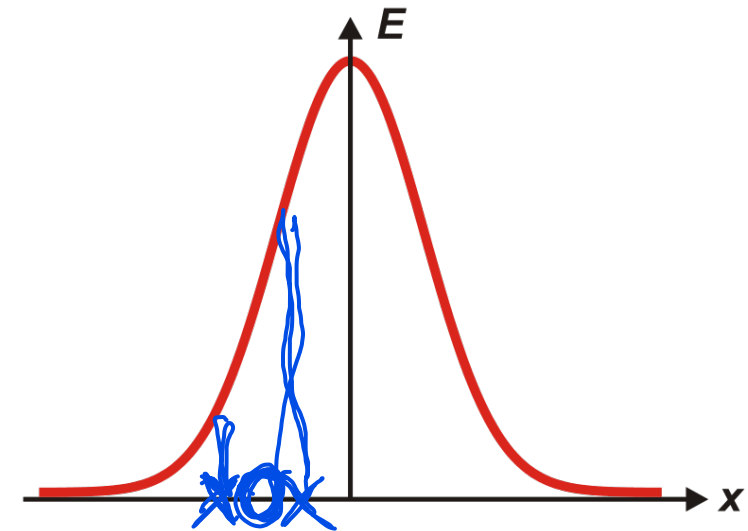
- Gaussian distribution in R

$$p(x|\mu, \sigma) = \frac{1}{(2\pi)^{\frac{1}{2}}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right)$$

- Need to estimate two sets of parameters μ, σ

- Given m iid samples

$$\mathcal{D} = \{x^1, x^2, \dots, x^m\}, x^i \in R$$



- Likelihood of one data point:

$$p(x^i|\mu, \sigma) \propto \exp\left(-\frac{1}{2\sigma^2}(x^i - \mu)^2\right)$$

The estimators for μ, σ are well-known

- Gaussian distribution in R

$$p(x|\mu, \sigma) = \frac{1}{(2\pi)^{\frac{1}{2}}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right)$$

$\vec{x}$

$$P(\vec{x} | \vec{\mu}, \Sigma) = \frac{1}{(2\pi)^{\frac{1}{2}} |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{(\vec{x} - \vec{\mu})^T \Sigma^{-1} (\vec{x} - \vec{\mu})}{2}\right)$$

- Mean

$$\mu = \frac{1}{m} \sum_{i=1}^m x^i$$

$$\vec{\mu} = \frac{1}{m} \sum_{i=1}^m \vec{x}^i$$

- Variance

$$\sigma^2 = \frac{1}{m} \sum_{i=1}^m (x^i - \mu)^2$$

$$\Sigma = \frac{1}{m} \sum_{i=1}^m (\vec{x}^i - \vec{\mu})(\vec{x}^i - \vec{\mu})^T$$

MLE for Gaussian distribution

- Objective function, log likelihood

$$\begin{aligned} l(\mu, \sigma; \mathcal{D}) &= \log \prod_{i=1}^m \frac{1}{(2\pi)^{\frac{1}{2}} \sigma} \exp \left(-\frac{1}{2\sigma^2} (x^i - \mu)^2 \right) \\ &= -\frac{m}{2} \log 2\pi - \frac{m}{2} \log \sigma^2 - \sum_{i=1}^m \frac{(x^i - \mu)^2}{2\sigma^2} \end{aligned}$$

- Maximize $l(\mu, \sigma; \mathcal{D})$ with respect to μ, σ
- Take derivatives w.r.t. μ, σ^2

$$\begin{aligned} \frac{\partial l}{\partial \mu} &= 0 \\ \frac{\partial l}{\partial \sigma^2} &= 0 \end{aligned}$$

$$l(\mu, \sigma; \mathcal{D}) = -\frac{m}{2} \log 2\pi - \frac{m}{2} \log \sigma^2 - \sum_{i=1}^m \frac{(x^i - \mu)^2}{2\sigma^2}$$

$$\frac{\partial l}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^m (x^i - \mu) = 0$$

$$\Rightarrow \sum_i x^i = m\mu \Rightarrow \mu = \frac{1}{m} \sum_{i=1}^m x^i$$

Handwritten notes:

$$\sigma^2 = \frac{1}{m-1} \sum (x^i - \mu)^2$$

~~MLE~~

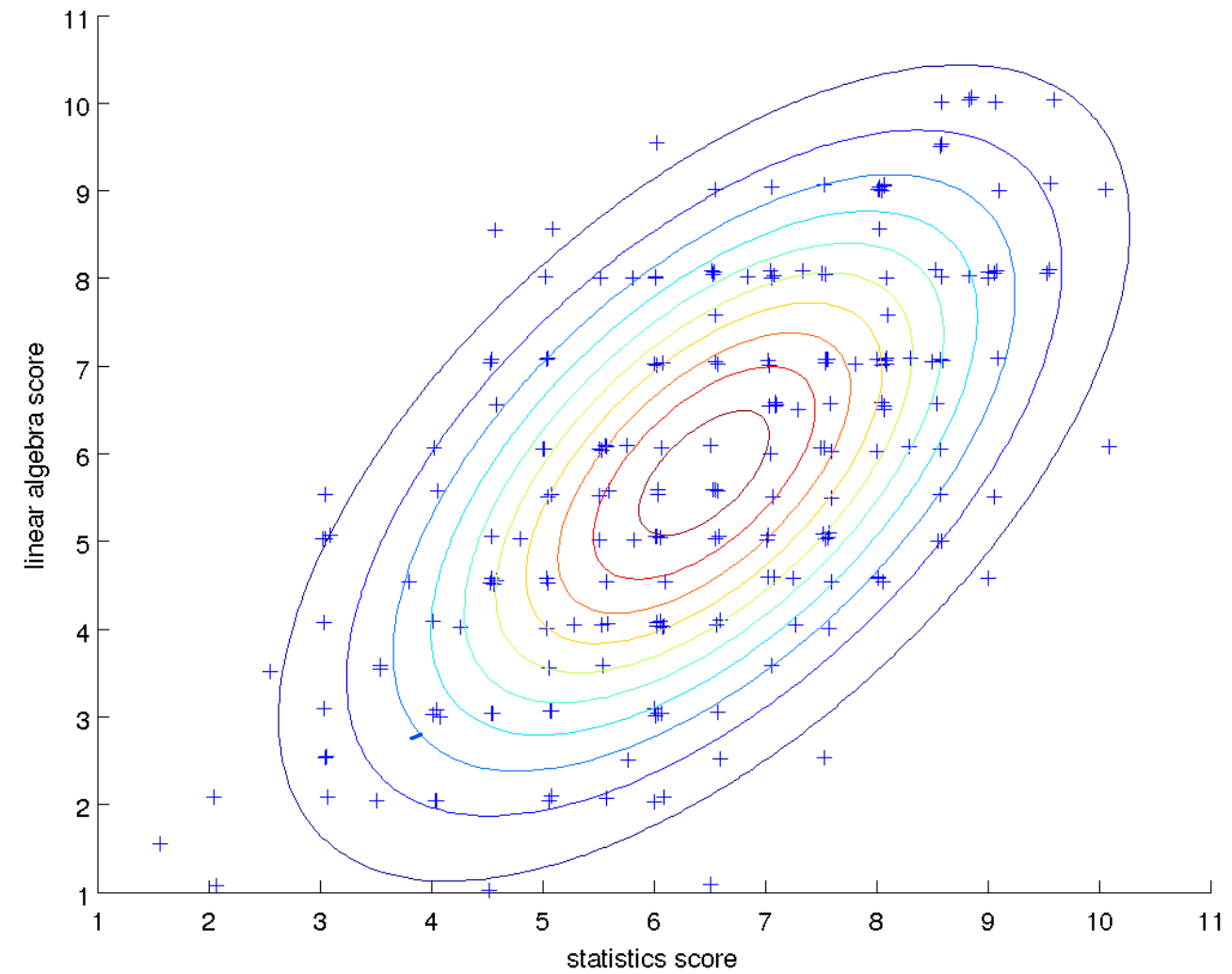
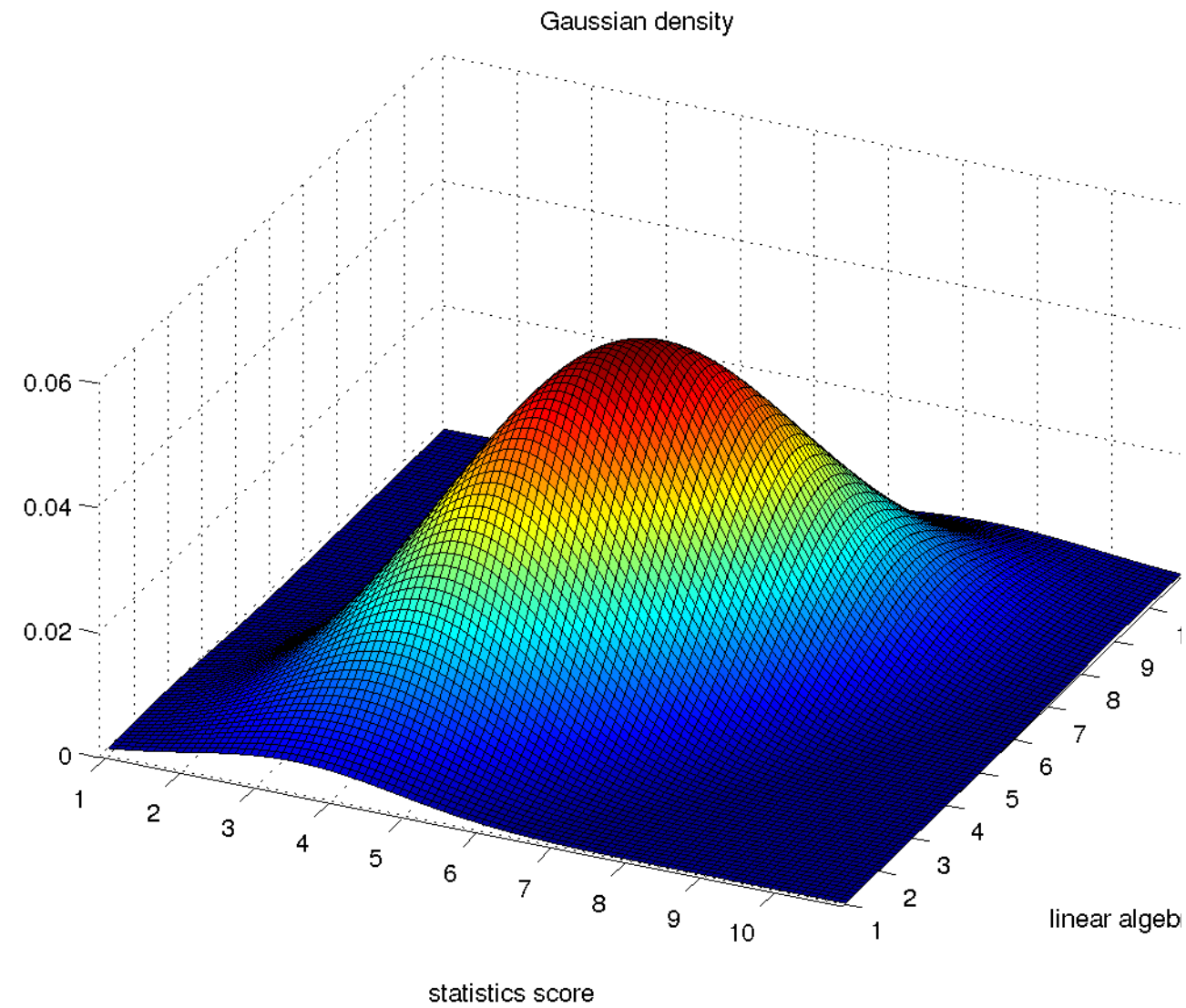
$$\frac{\partial l}{\partial \sigma^2} = -\frac{m}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_i (x^i - \mu)^2 = 0$$

$$\Rightarrow \sum_i (x^i - \mu)^2 = m\sigma^2 \Rightarrow \frac{1}{m} \sum_{i=1}^m (x^i - \mu)^2$$

Density example

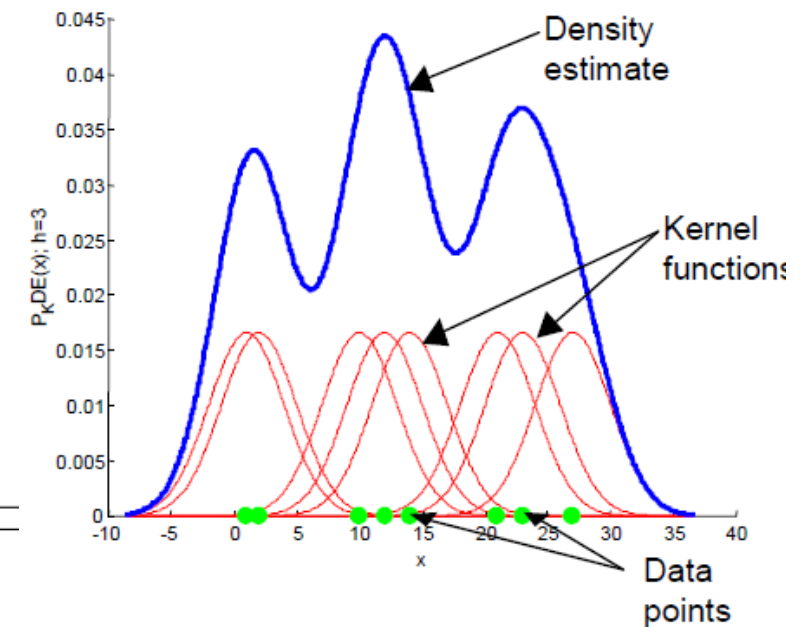
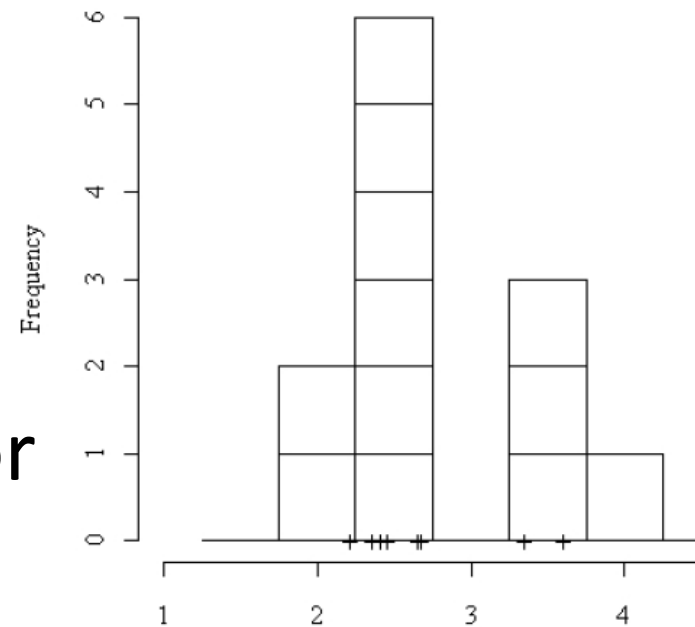
$$\Sigma = \begin{bmatrix} 4 & -2 \\ -2 & 3.5 \end{bmatrix}$$

statistics vs linear algebra (Gaussian)



Nonparametric models

- Eg. Histogram
- Eg. Kernel density estimator



- What are nonparametric models?
 - “nonparametric” does **not** mean there are no parameters
 - can not be described by a fixed number of parameters
 - one can think of there are many many parameters

1-D Histogram

- One the simplest nonparametric density estimator

- Given m iid samples $\mathcal{D} = \{x^1, x^2, \dots, x^m\}, x^i \in [0,1)$

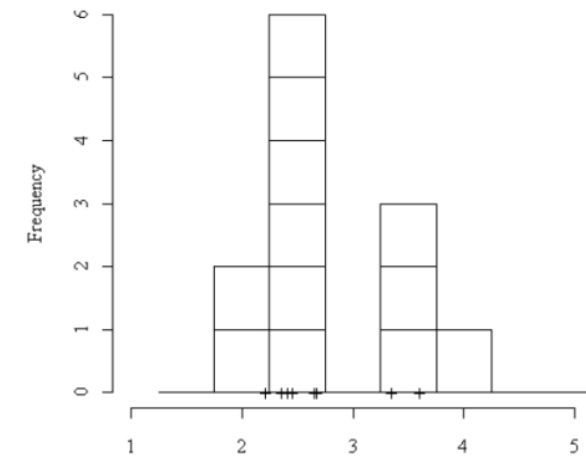
- Split $[0,1)$ into n bins

$$B_1 = \left[0, \frac{1}{n}\right), B_2 = \left[\frac{1}{n}, \frac{2}{n}\right), \dots, B_n = \left[\frac{n-1}{n}, 1\right)$$

- Count the number of points, c_1 within B_1 , c_2 within B_2 ...

- For a new test point x

$$p(x) = \sum_{j=1}^n \frac{nc_j}{m} I(x \in B_j)$$



Why is histogram valid?

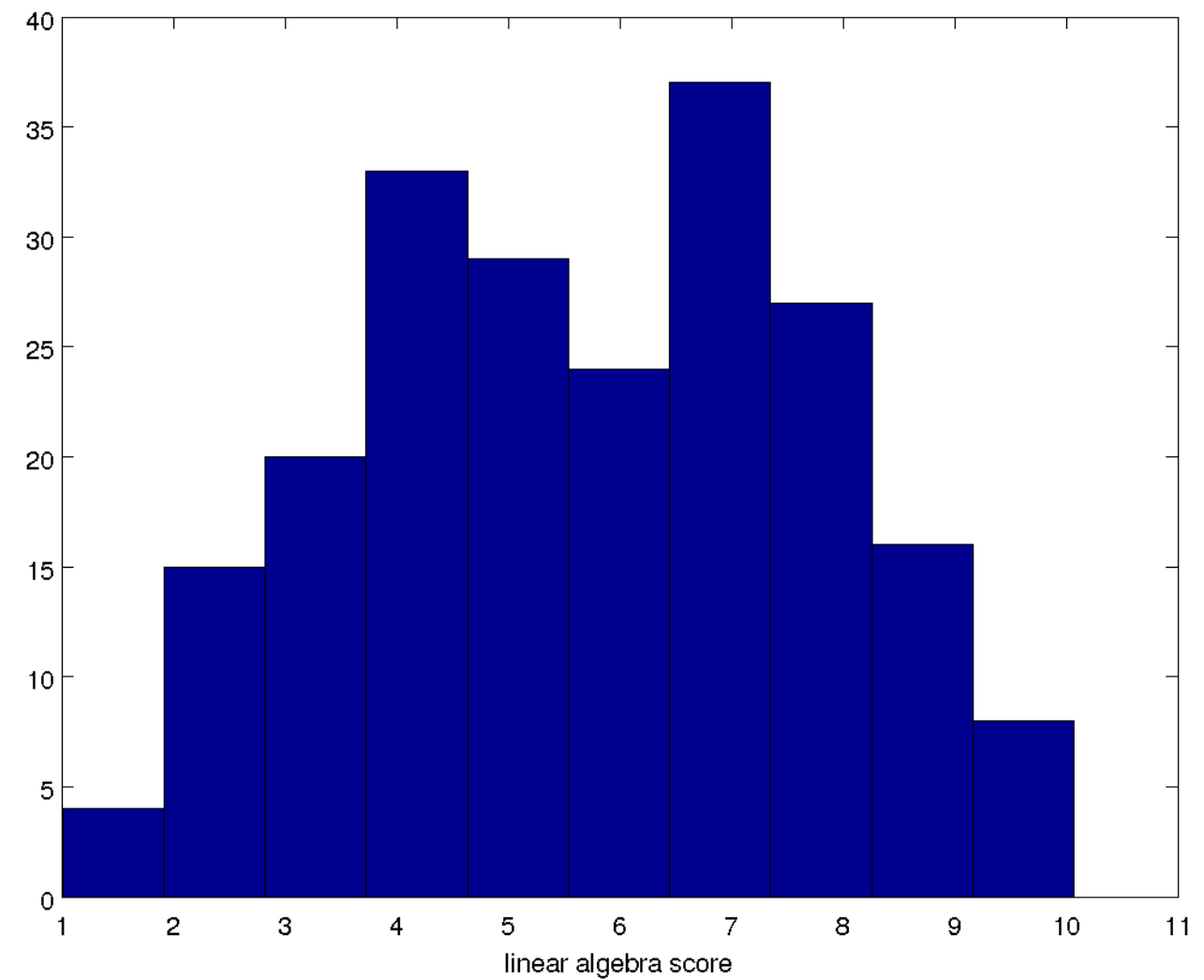
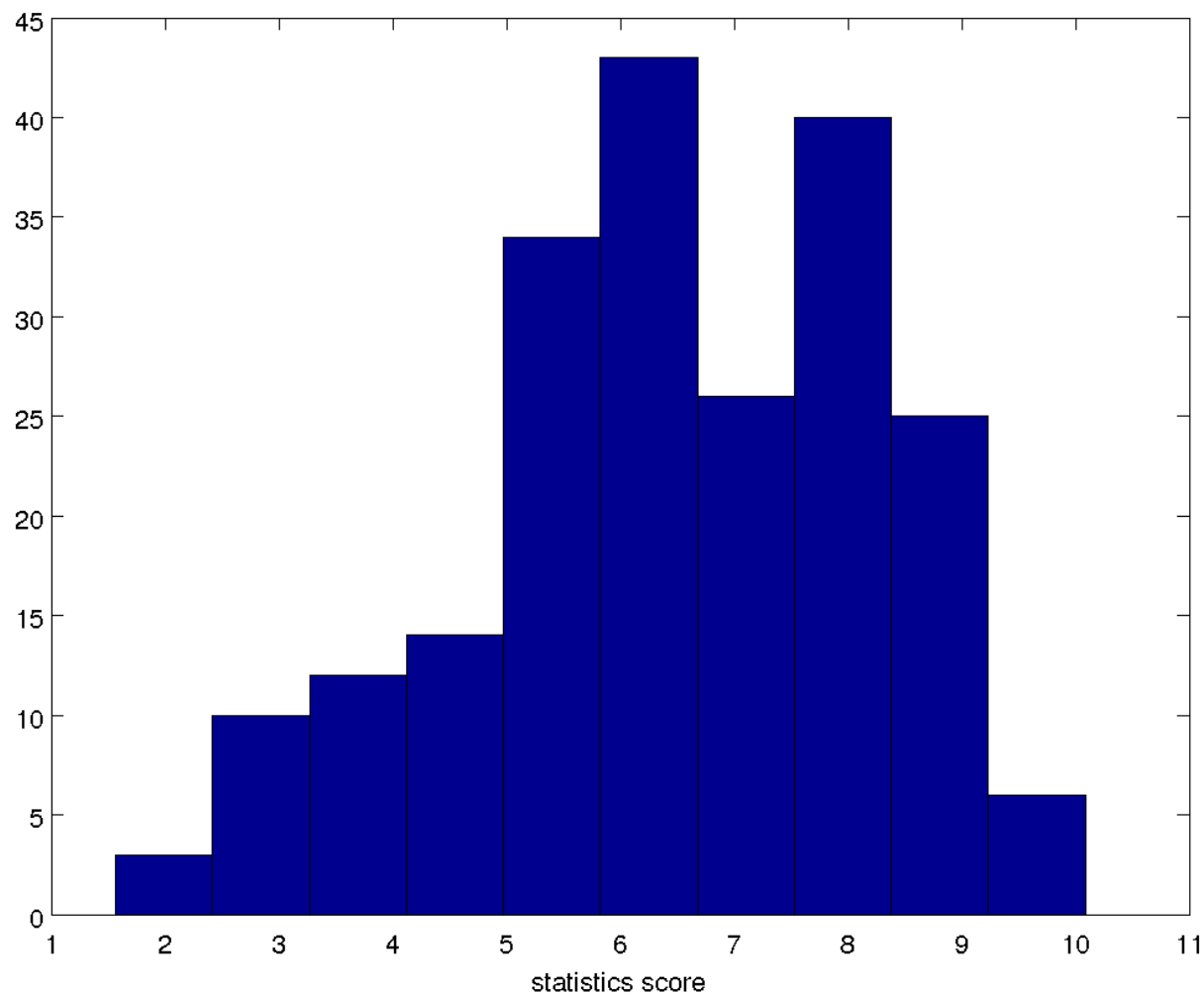
- Requirement for density $p(x)$
- $p(x) \geq 0, \int_{\Omega} p(x)dx = 1$

- For histogram,

$$\begin{aligned}\int_{[0,1)} p(x)dx &= \int_{[0,1)} \sum_{j=1}^n \frac{nc_j}{m} I(x \in B_j) dx \\ &= \sum_{j=1}^n \int_{\left[\frac{j-1}{n}, \frac{j}{n}\right)} \frac{nc_j}{m} dx \\ &= \sum_{j=1}^n \frac{c_j}{m} = 1\end{aligned}$$

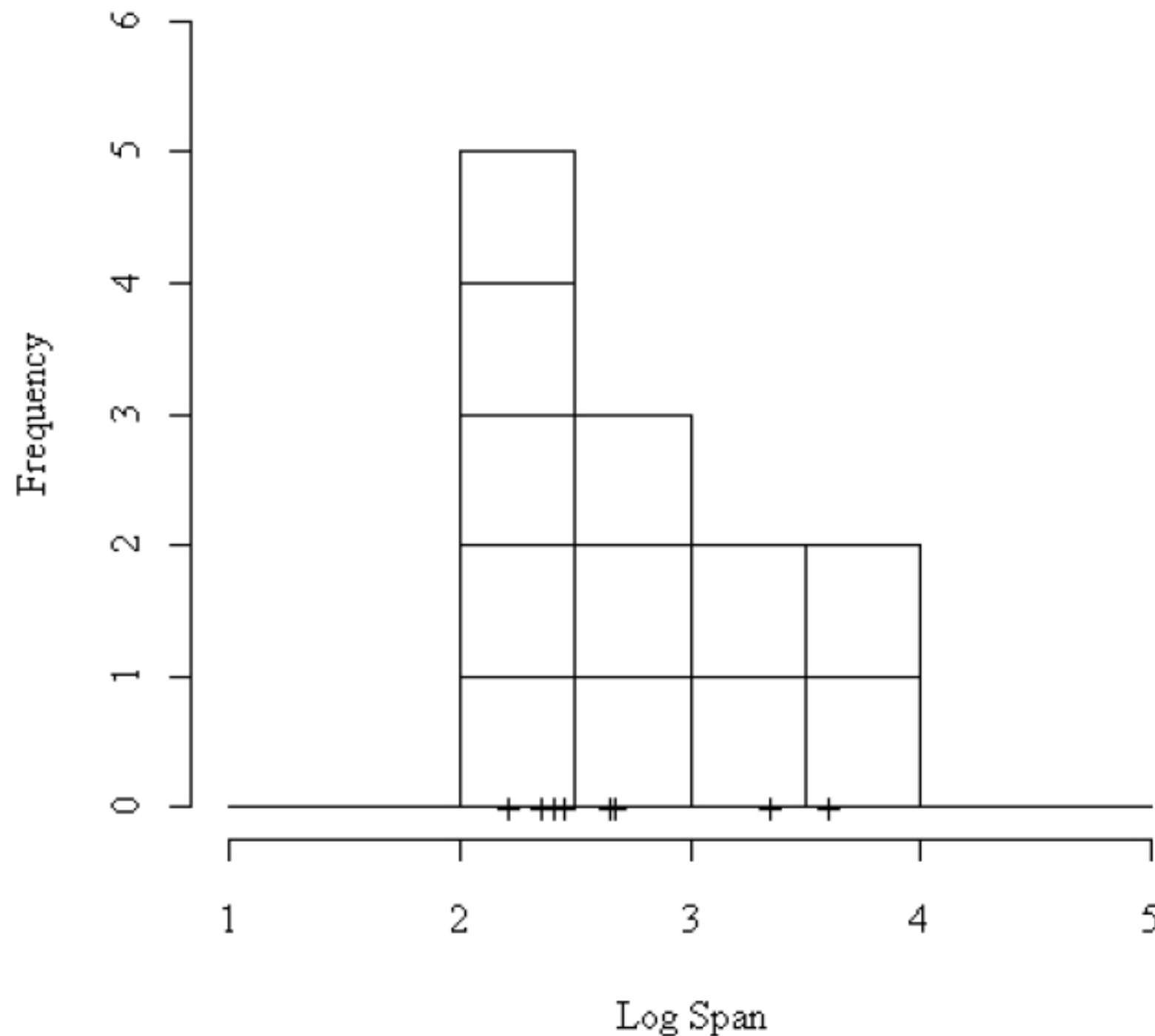
Example: test scores (cont.)

- What is missing if we want density?



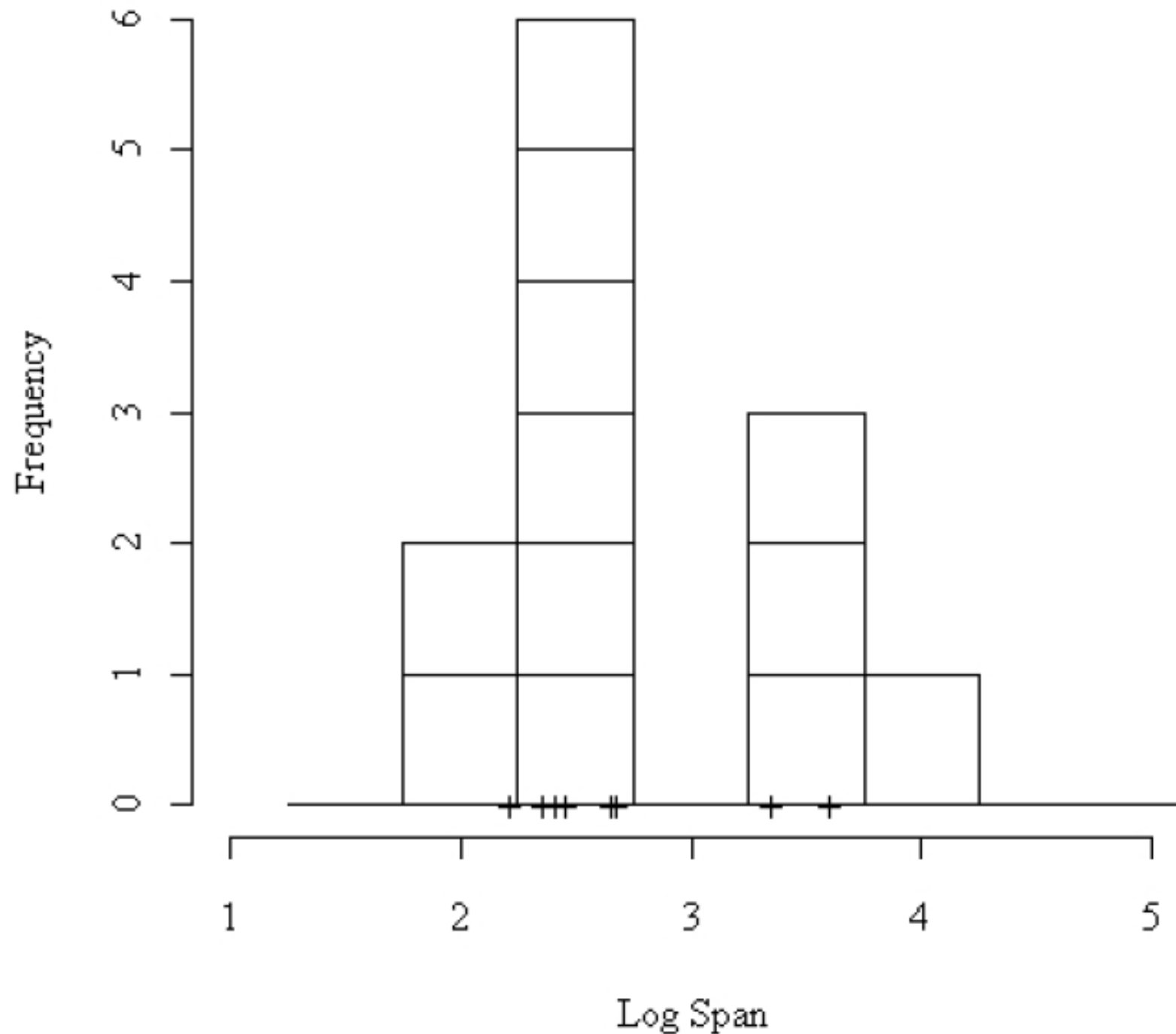
Output depends on where you put the bins

**Histogram with breaks at n.0 and n.5
binwidth=0.5**



Output depends on where you put the bins

**Histogram with breaks at n.25 and n.75
binwidth=0.5**



Higher dimensional histogram

- Given m iid samples $\mathcal{D} = \{x^1, x^2, \dots, x^m\}$, $x^i \in [0,1)^d$

- Split $[0,1)^d$ evenly into n^d bins

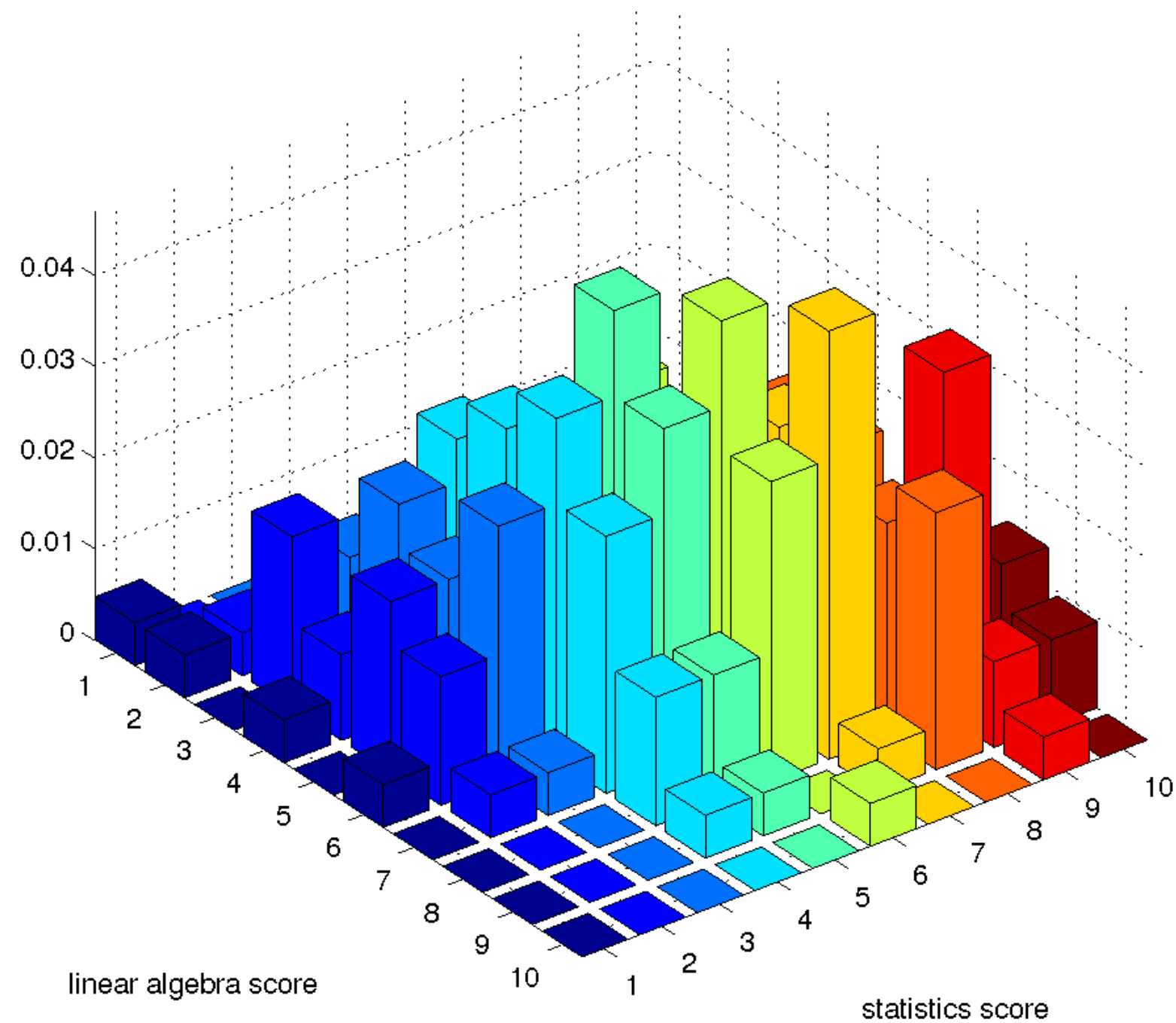
$$B_1 = \left[0, \frac{1}{n}\right) \times \left[0, \frac{1}{n}\right) \dots \times \left[0, \frac{1}{n}\right),$$

$$B_2 = \left[\frac{1}{n}, \frac{2}{n}\right) \times \left[0, \frac{1}{n}\right) \dots \times \left[0, \frac{1}{n}\right),$$

$$\dots$$
$$B_{n^d} = \left[\frac{n-1}{n}, 1\right) \times \left[\frac{n-1}{n}, 1\right) \dots \times \left[\frac{n-1}{n}, 1\right)$$

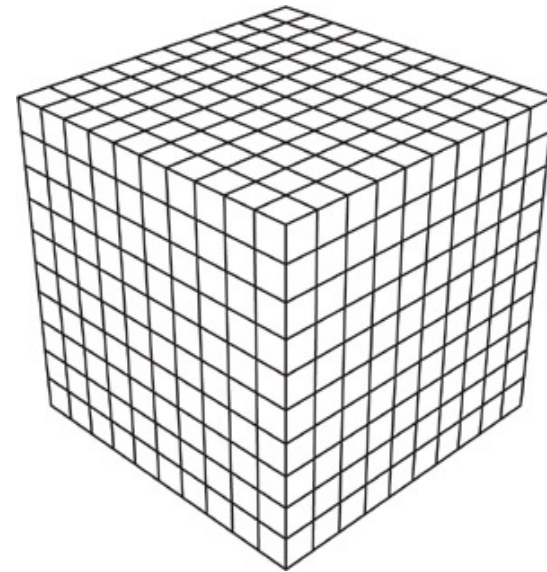
- Bin size is $h = \frac{1}{n}$

Class scores



Computation and statistical considerations

- Problem I: **too many bin! Not good for high dimensional data**
 - If n^d is larger than m , most bins are empty
 - Eg. $n = 10, d = 6$, need ~ 1 million data points



- Problem II: statistically histogram is not the best

- Integrated risk:

$$r(\hat{p}, p) := \int_R \mathbb{E}_X \left[(\hat{p}(x) - p(x))^2 \right] dx$$

- Histogram (with bin size $h \sim m^{-1/3}$)

$$r(\hat{p}, p) \sim \frac{C}{m^{2/3}}$$

- Kernel density estimator (with bandwidth $h \sim m^{-1/5}$)

$$r(\hat{p}, p) \sim \frac{C}{m^{4/5}}$$

- Difference even big for high dimensional data

Kernel density estimation

- Kernel density estimator

$$p(x) = \frac{1}{m} \sum_i^m \frac{1}{h} K\left(\frac{x^i - x}{h}\right)$$

- Smoothing kernel function

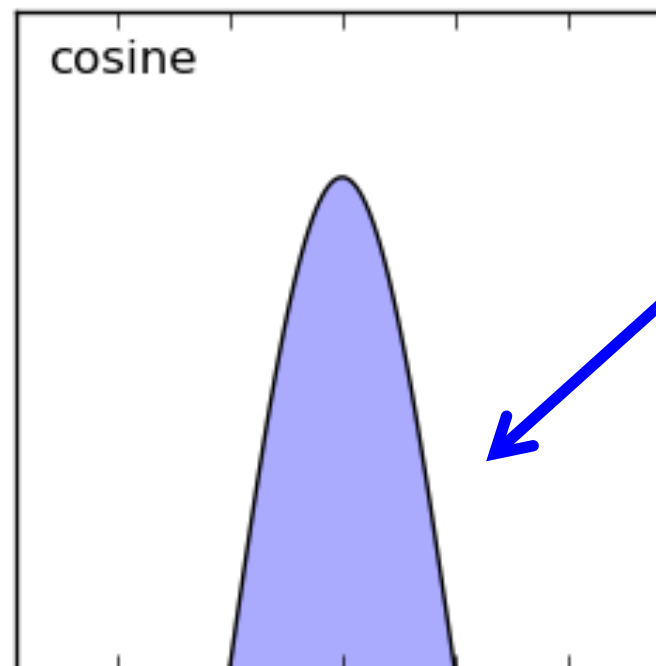
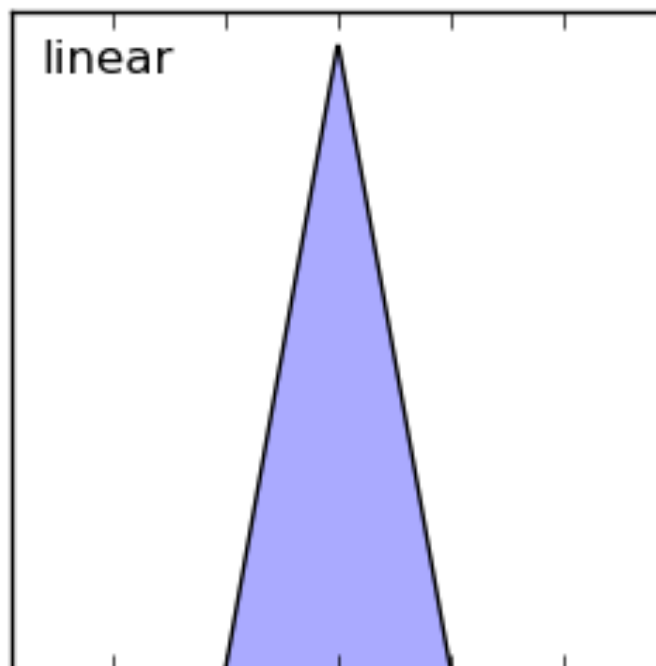
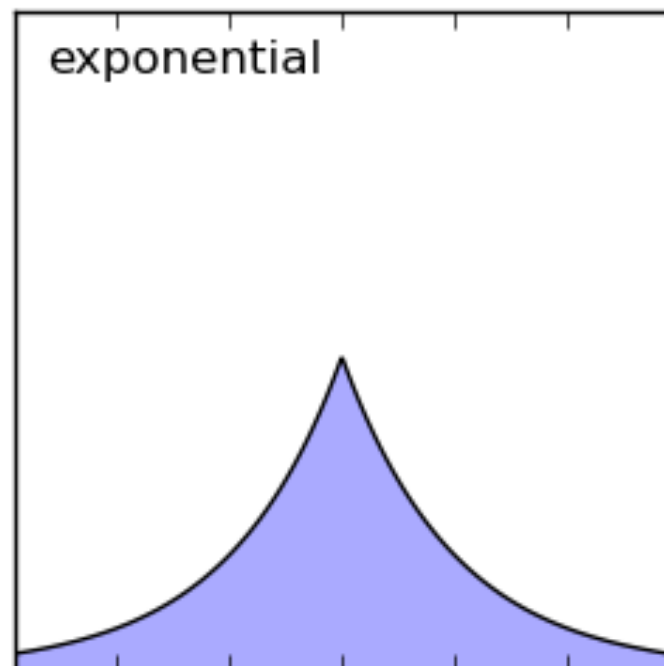
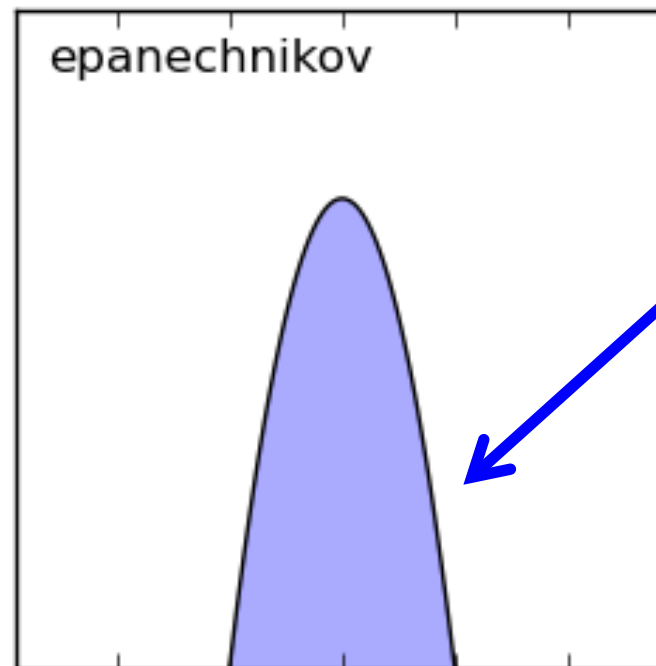
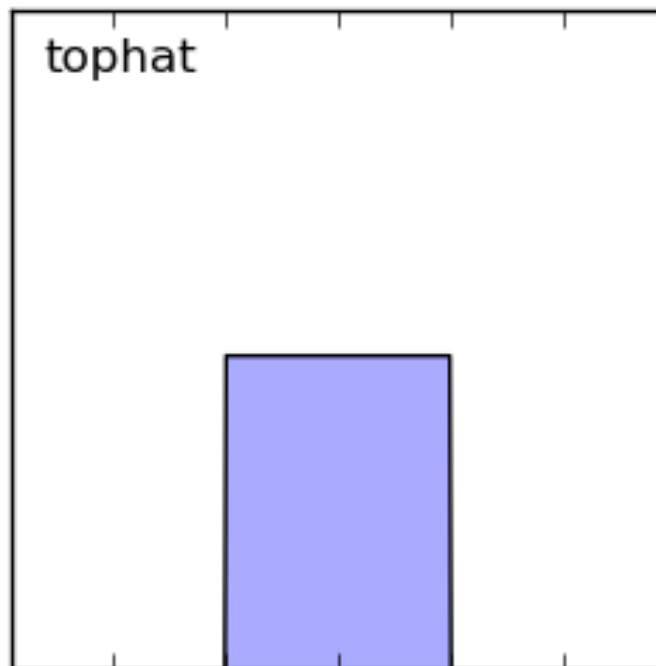
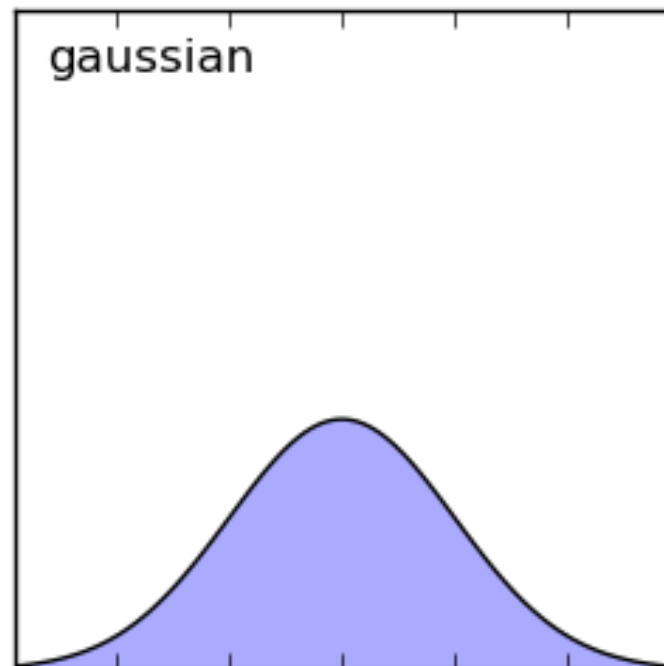
- $K(u) \geq 0$,
- $\int K(u)du = 1$,
- $\int uK(u) = 0$,
- $\int u^2 K(u)du < \infty$

- An example: Gaussian kernel $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$

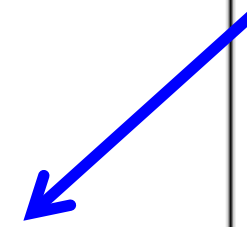
Smoothing kernel functions

- An example: Gaussian kernel $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$

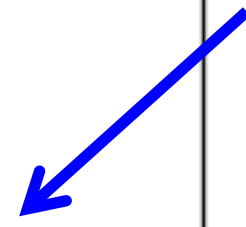
Available Kernels



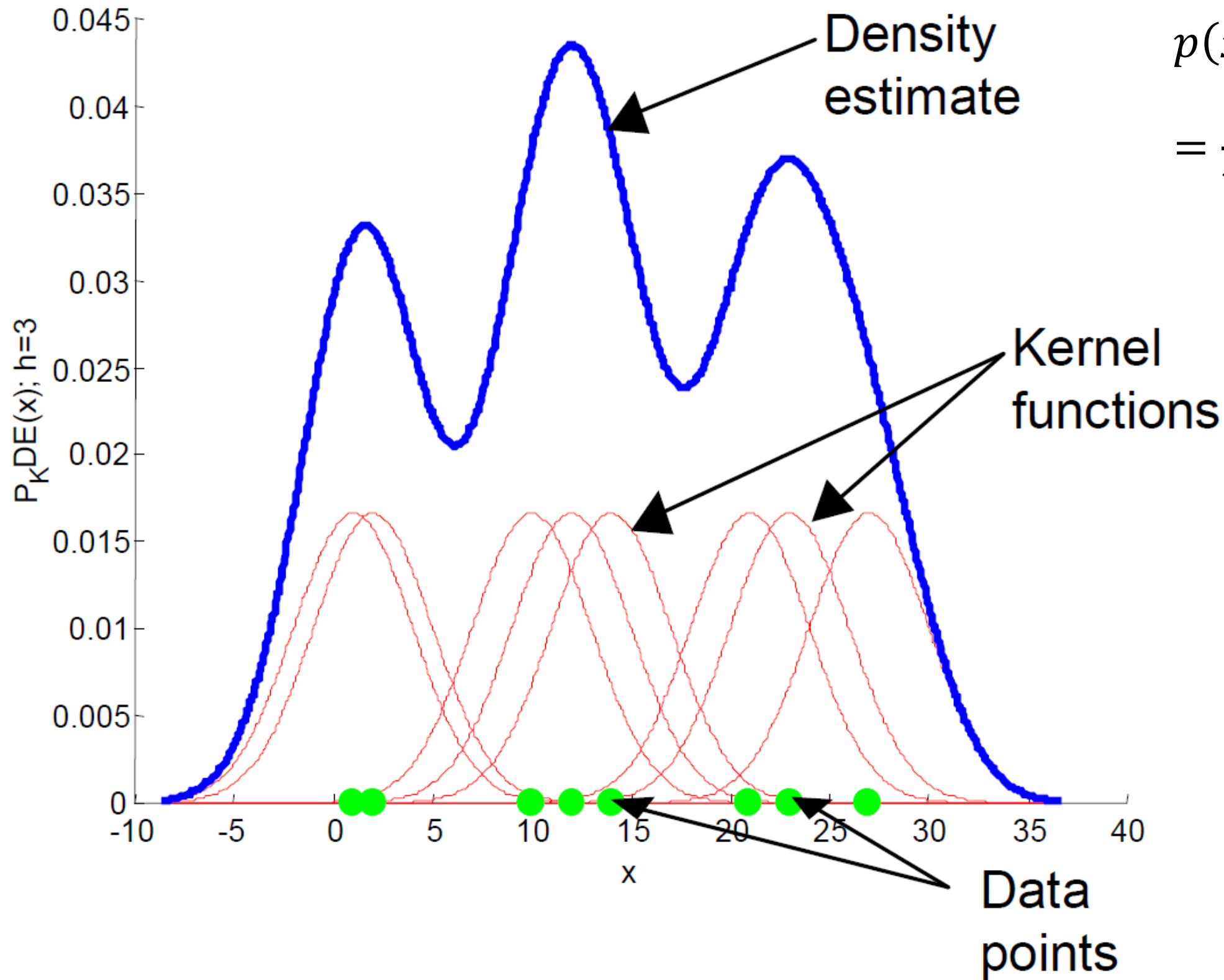
$$K(u) = \frac{3}{4}(1 - u^2)I(|u| \leq 1)$$



$$K(u) = \frac{\pi}{4} \cos\left(\frac{\pi}{2}u\right)I(|u| \leq 1)$$



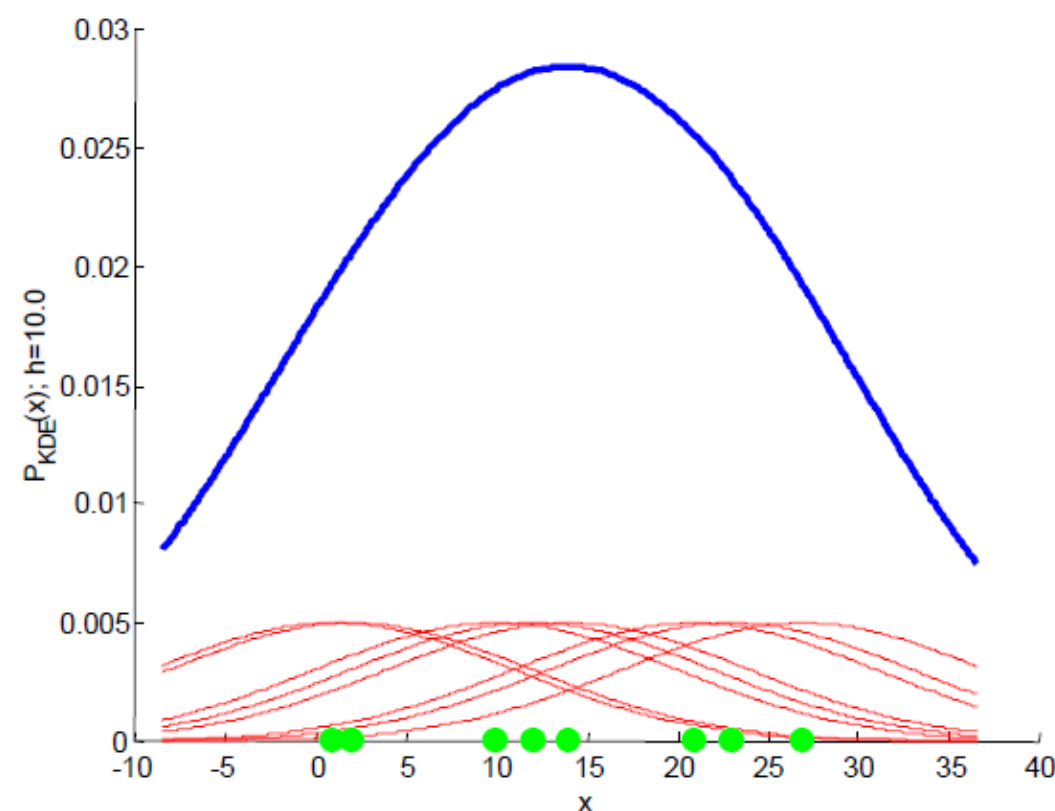
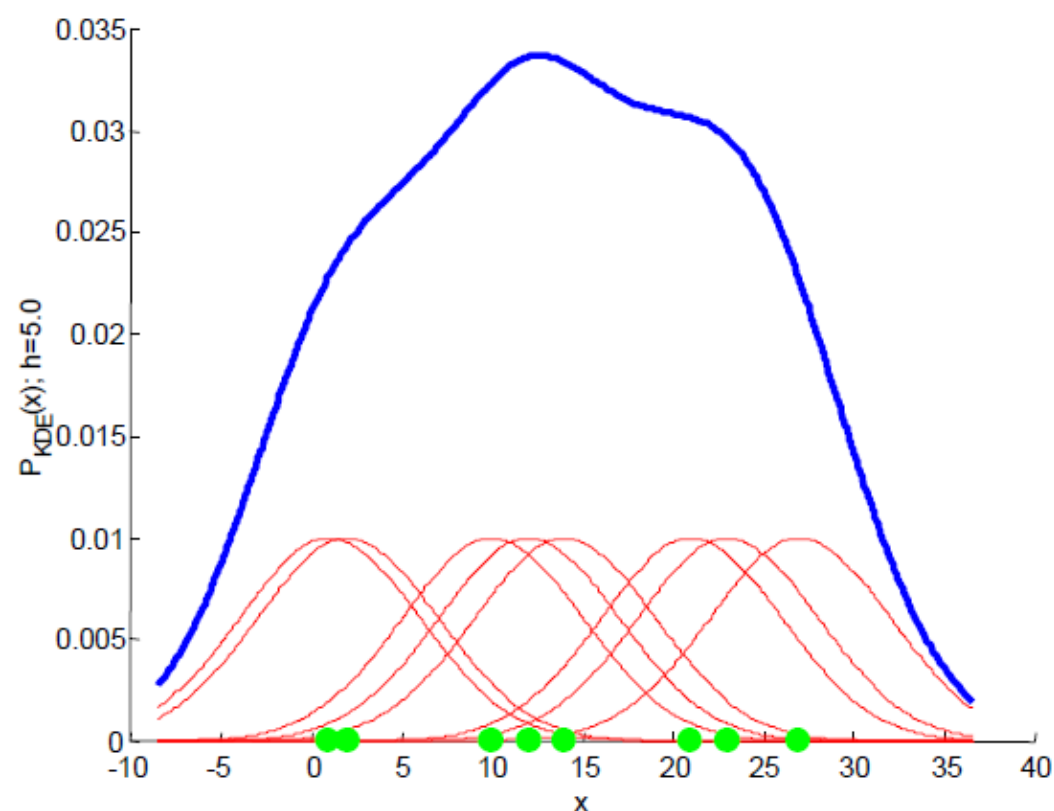
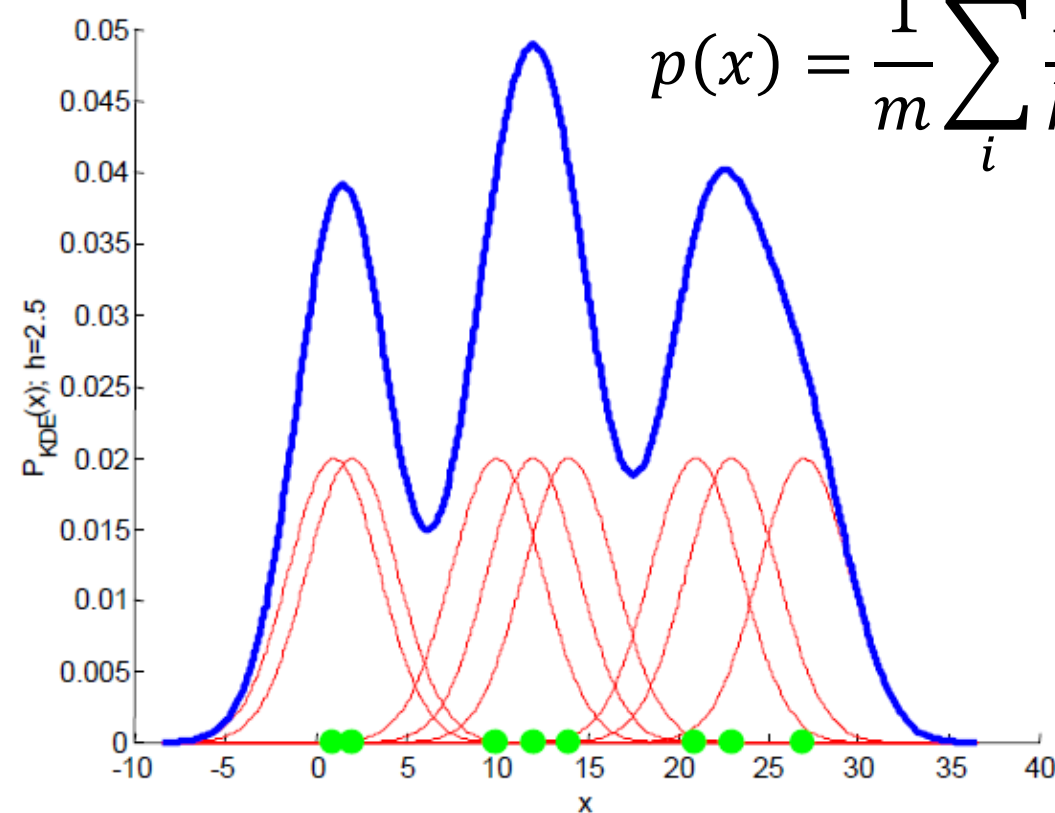
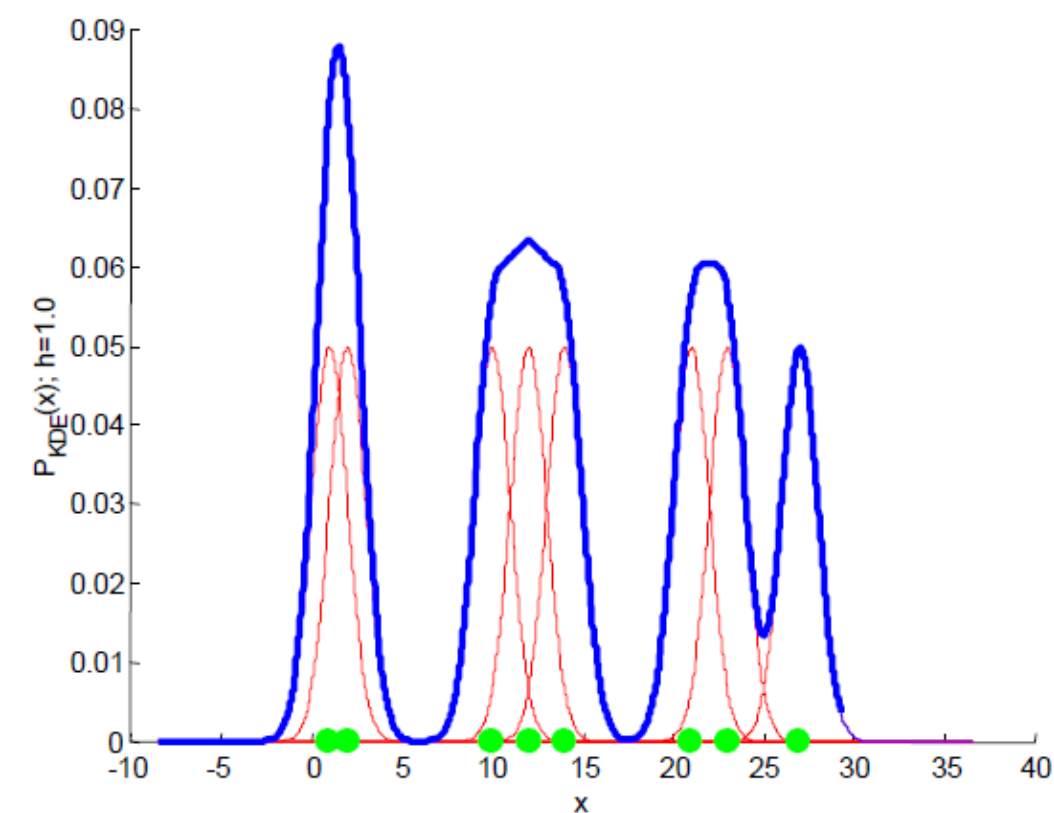
Example



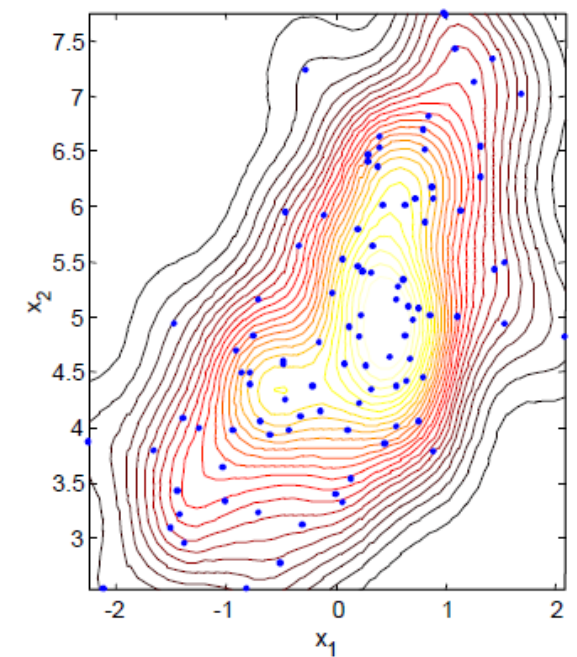
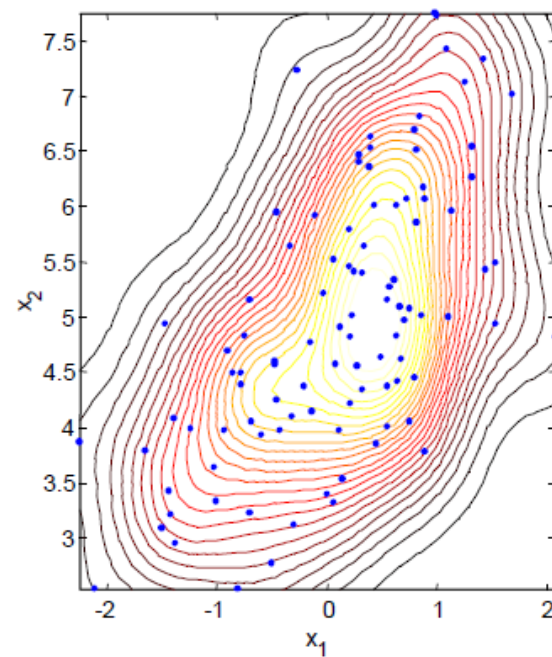
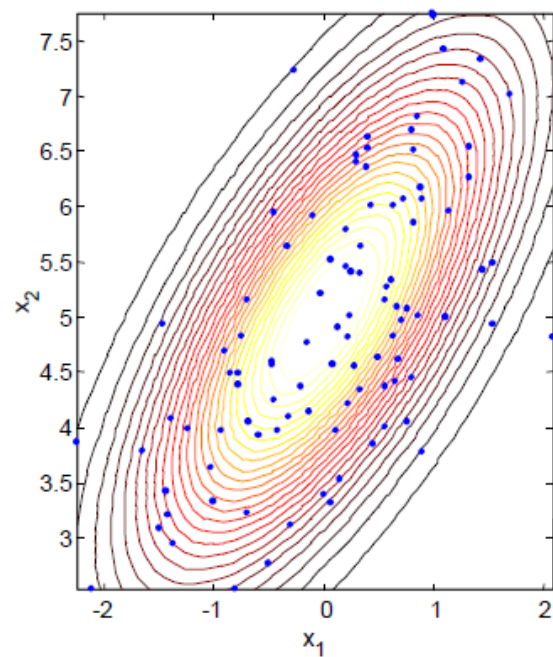
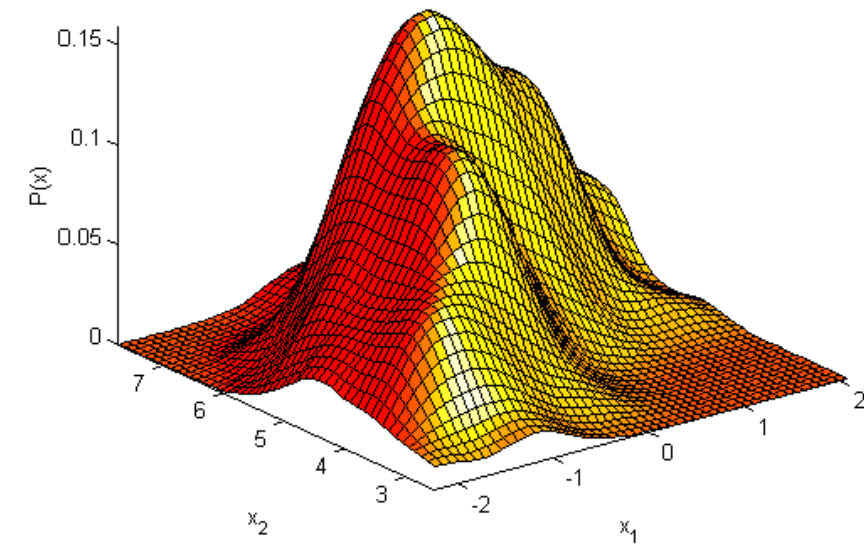
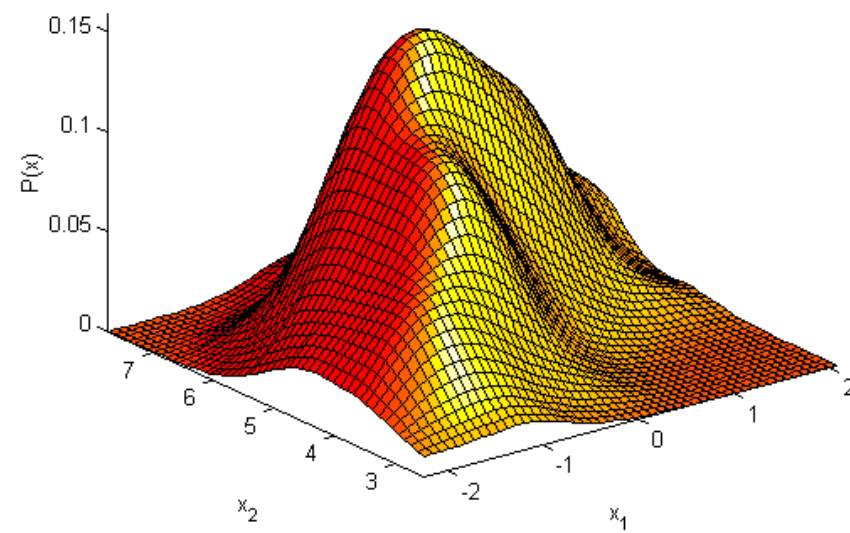
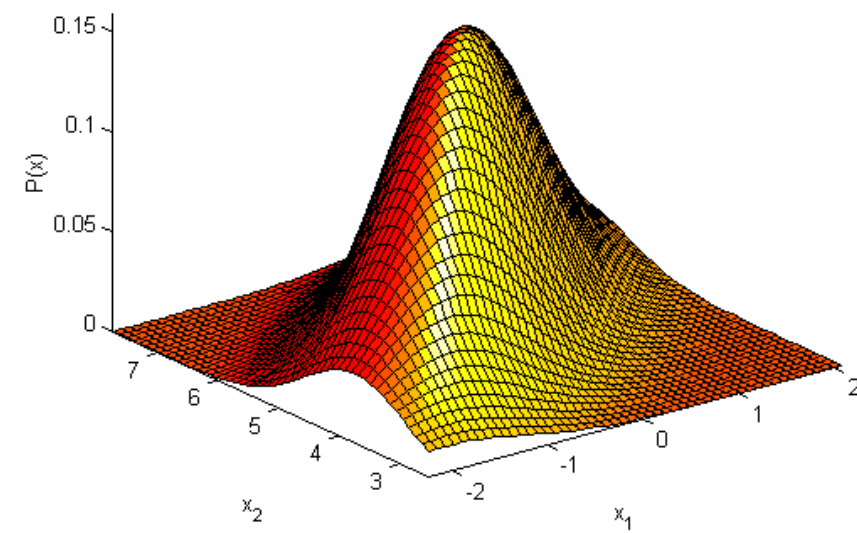
$$p(x) = \frac{1}{m} \sum_i^m \frac{1}{h} K\left(\frac{x^i - x}{h}\right)$$

Effect of the kernel bandwidth h

$$p(x) = \frac{1}{m} \sum_i^m \frac{1}{h} K \left(\frac{x^i - x}{h} \right)$$



Two-dimensional example



What is the best kernel bandwidth?

- Silverman's rule of thumb: If using the Gaussian kernel, a good choice for h is

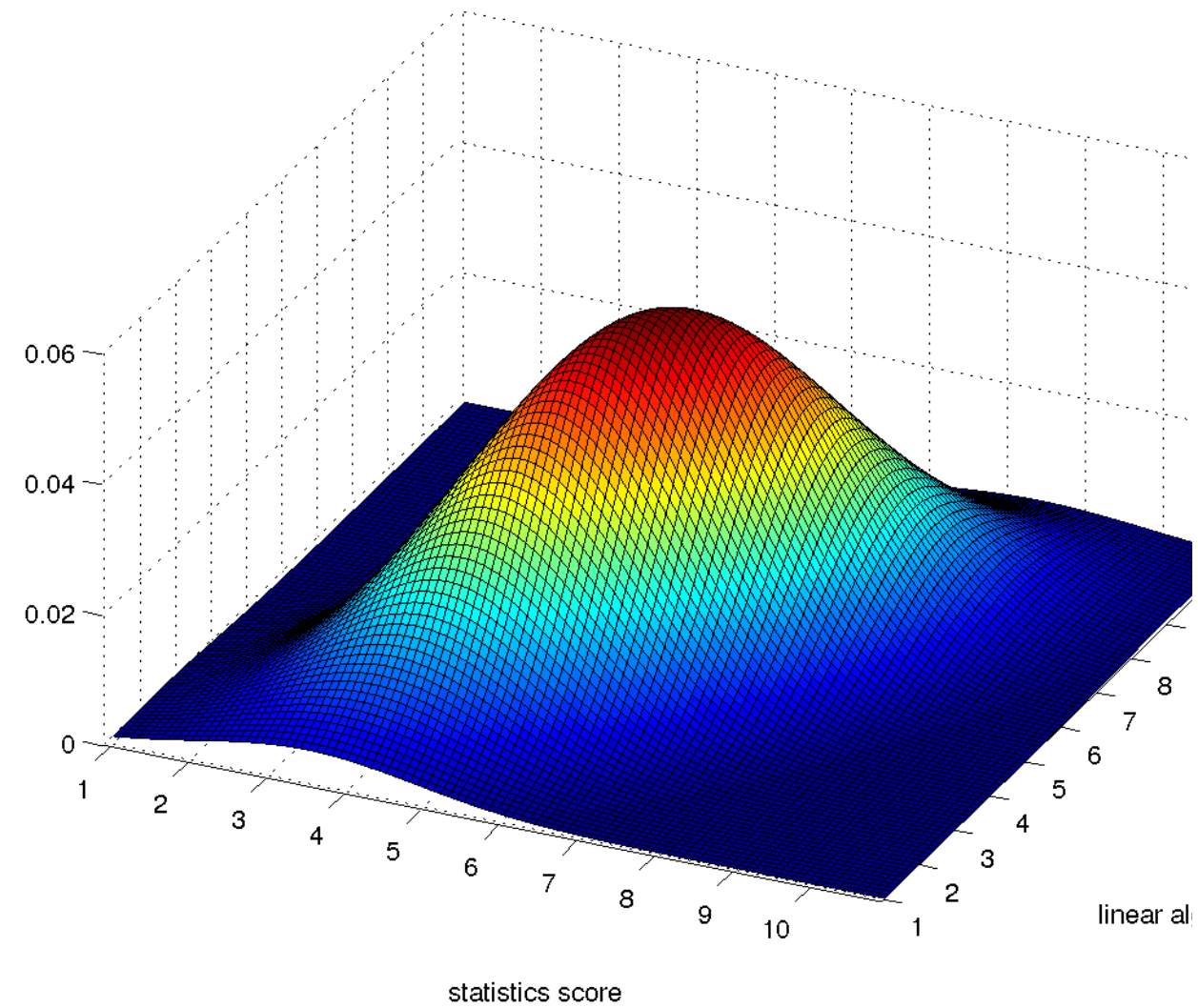
$$h \approx 1.06 \hat{\sigma} m^{-1/5}$$

where $\hat{\sigma}$ is the standard deviation of the samples

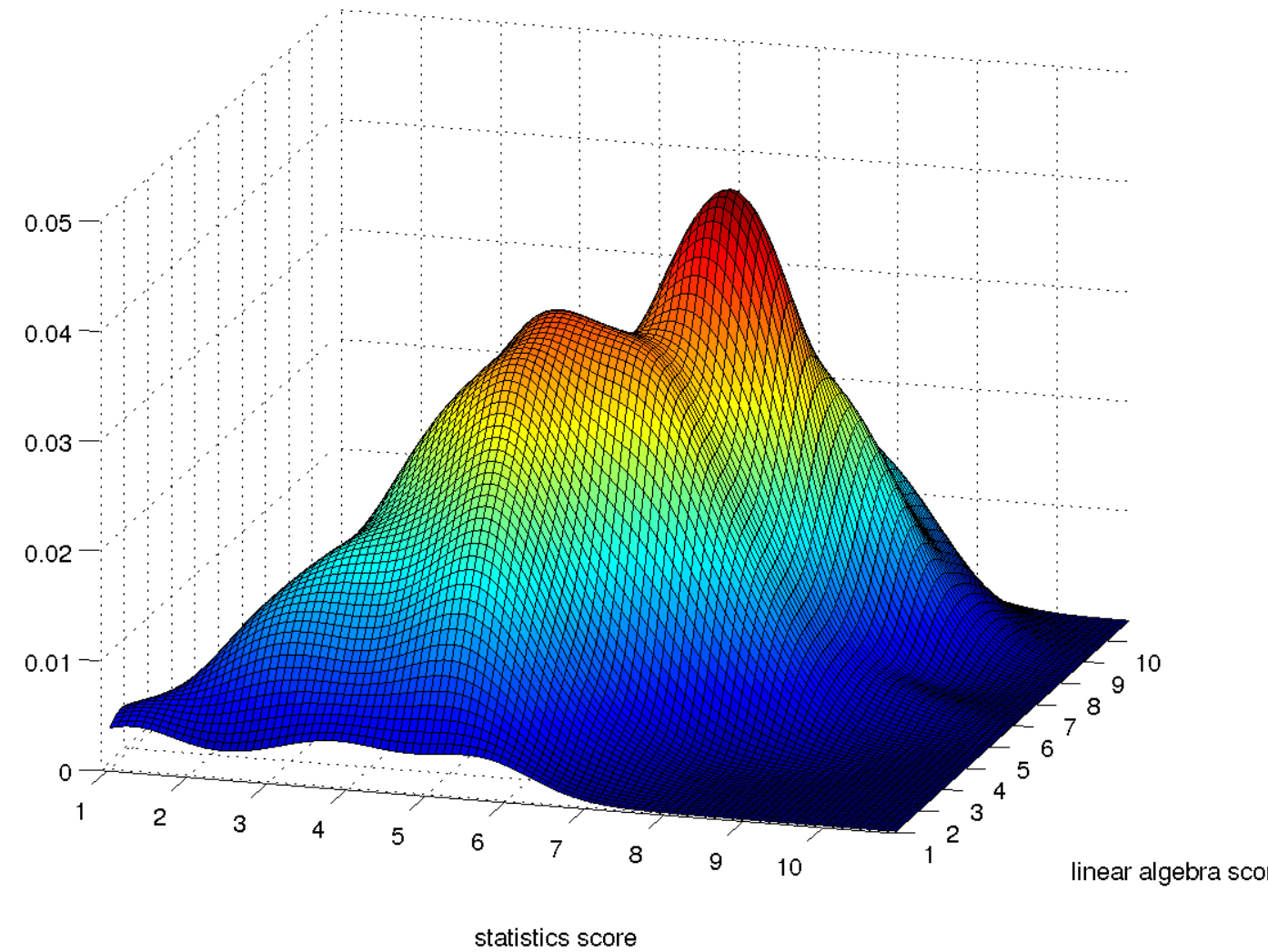
- A better but more computational intensive approach:
 - Randomly split the data into two sets
 - Obtain a kernel density estimate for the first
 - Measure the likelihood of the second set
 - Repeat over many random splits and average

Parametric vs. nonparametric

Gaussian density

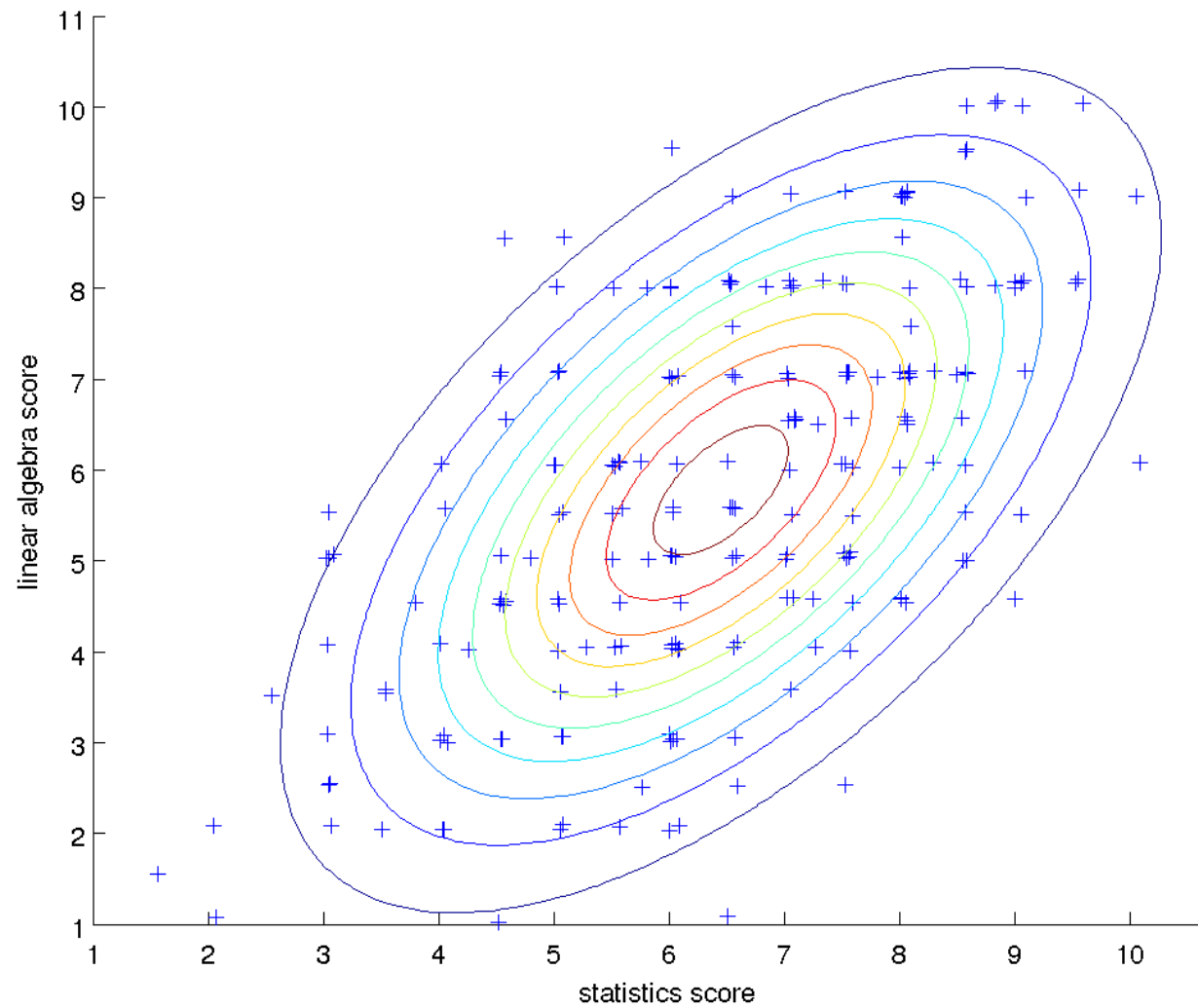


kernel density estimator

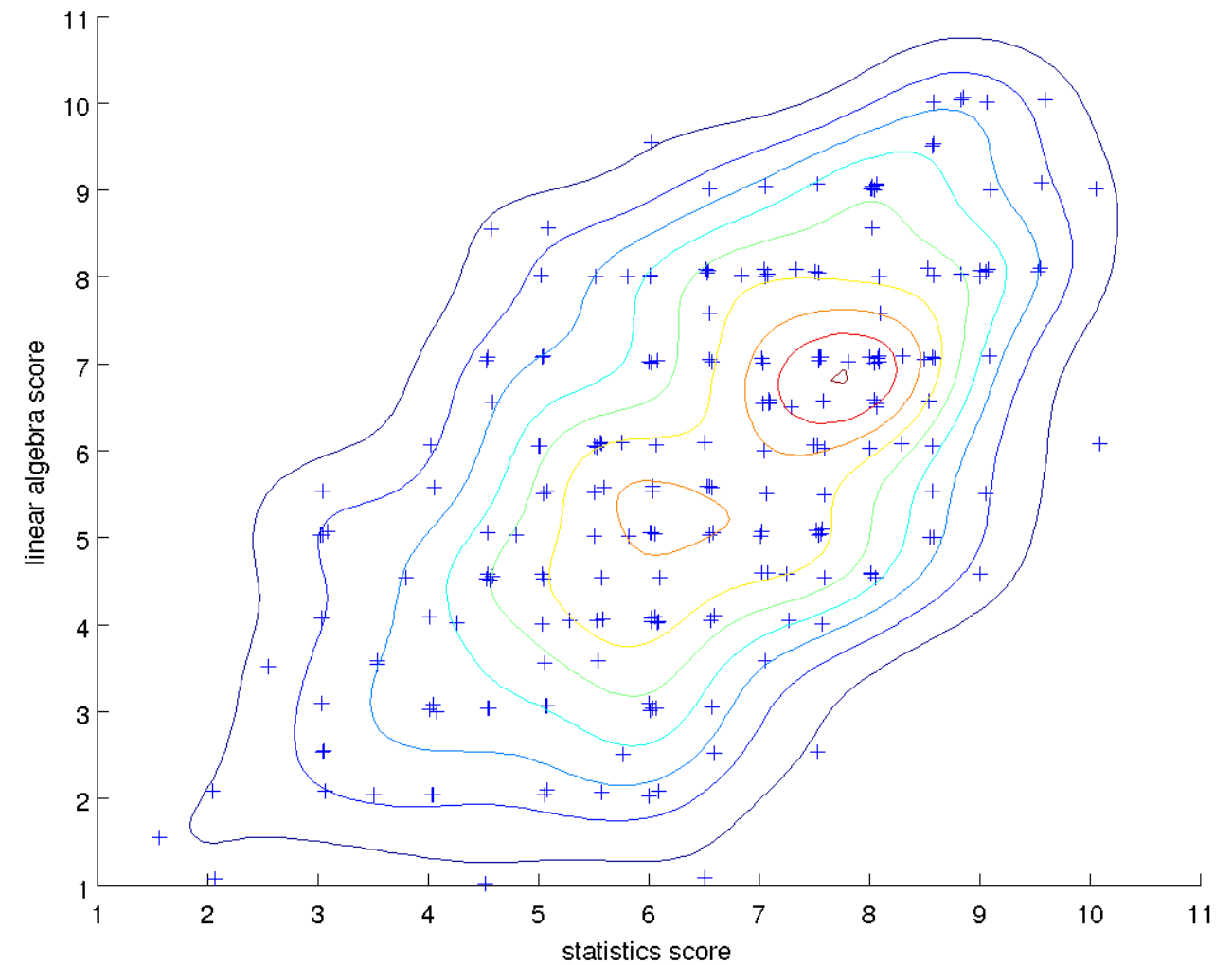


Parametric vs. nonparametric

statistics vs linear algebra (Gaussian)



statistics vs linear algebra



Parametric vs. nonparametric

- Nonparametric models place very mild assumptions on the data distribution and provide good models for complex data

Parametric models rely on very strong (simplistic) distributional assumptions

- Nonparametric models (not histograms) requires storing and computing with the entire data set.

Parametric models, once fitted, are much more efficient in terms of storage and computation.

Parametric vs. nonparametric

- Data $x \in R^d$ with **fixed** dimension d
- Given m training data points $\{x^1, x^2 \dots, x^m\}$
- Partition n bin in each dimension

Aspects	Gaussian	Histogram	KDE
Flexible	Not	Yes	Yes
Assumption	Strong	Not	Not
Parameter number	Fixed	Increase with n	Increase with m
Memory requirement	$d+d^2$	n^d	md
Training computation	Closed form	Binning and Counting	Nothing
Test computation	Plug in formula	Find the bin	Evaluate m functions
Statistical guarantee	Only Gaussian case	Arbitrary (worse)	Arbitrary (better)

Classification using density estimation

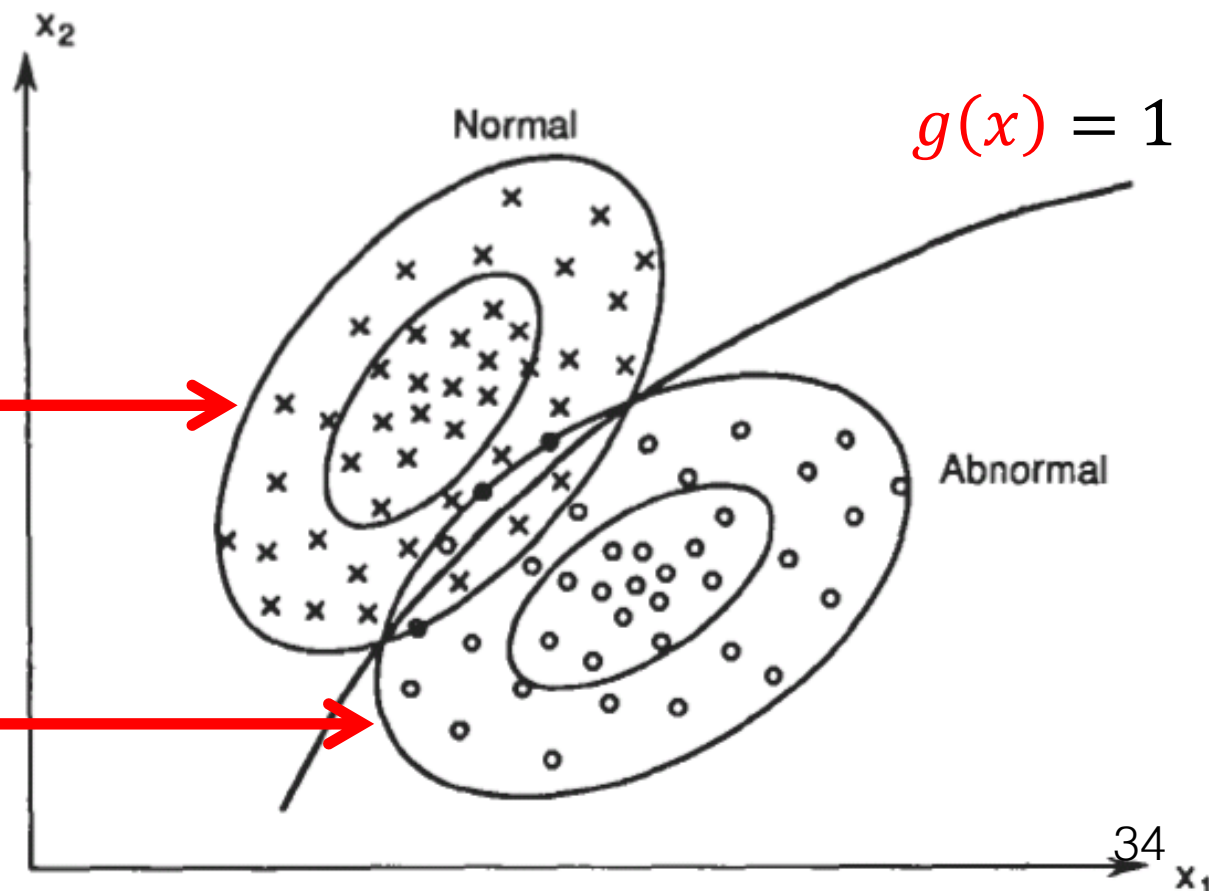
- Simple binary classifier for input $x^i \in R^d$ and label $y^i \in \{0,1\}$
 - Step I: use label to estimate $p(y = 0)$ and $p(y = 1)$
 - Step II: divide your data according to the value of y , and estimate $p(x|y = 0)$ and $p(x|y = 1)$

- Step III: Classify a new test point x as

$$g(x) := \begin{cases} 1, & \frac{p(x|y = 1)p(y = 1)}{p(x|y = 0)p(y = 0)} > 1 \\ 0, & \text{otherwise} \end{cases}$$

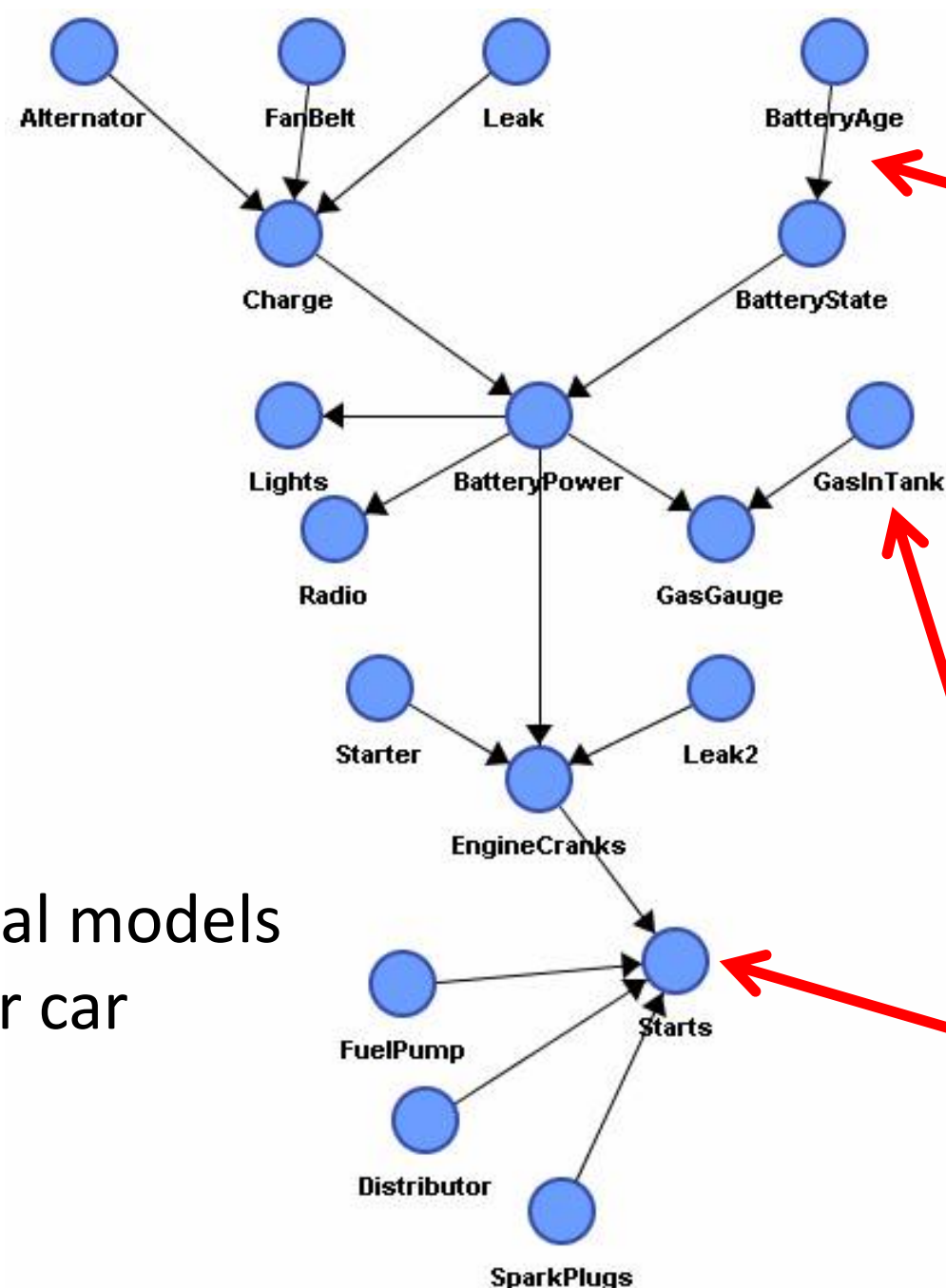
$p(x|y = 0)$

$p(x|y = 1)$



Graphical models with density estimation

- (Probabilistic) graphical models
 - Use graph to describe the relation of many variables
 - Reason about some unknown variable given known variables



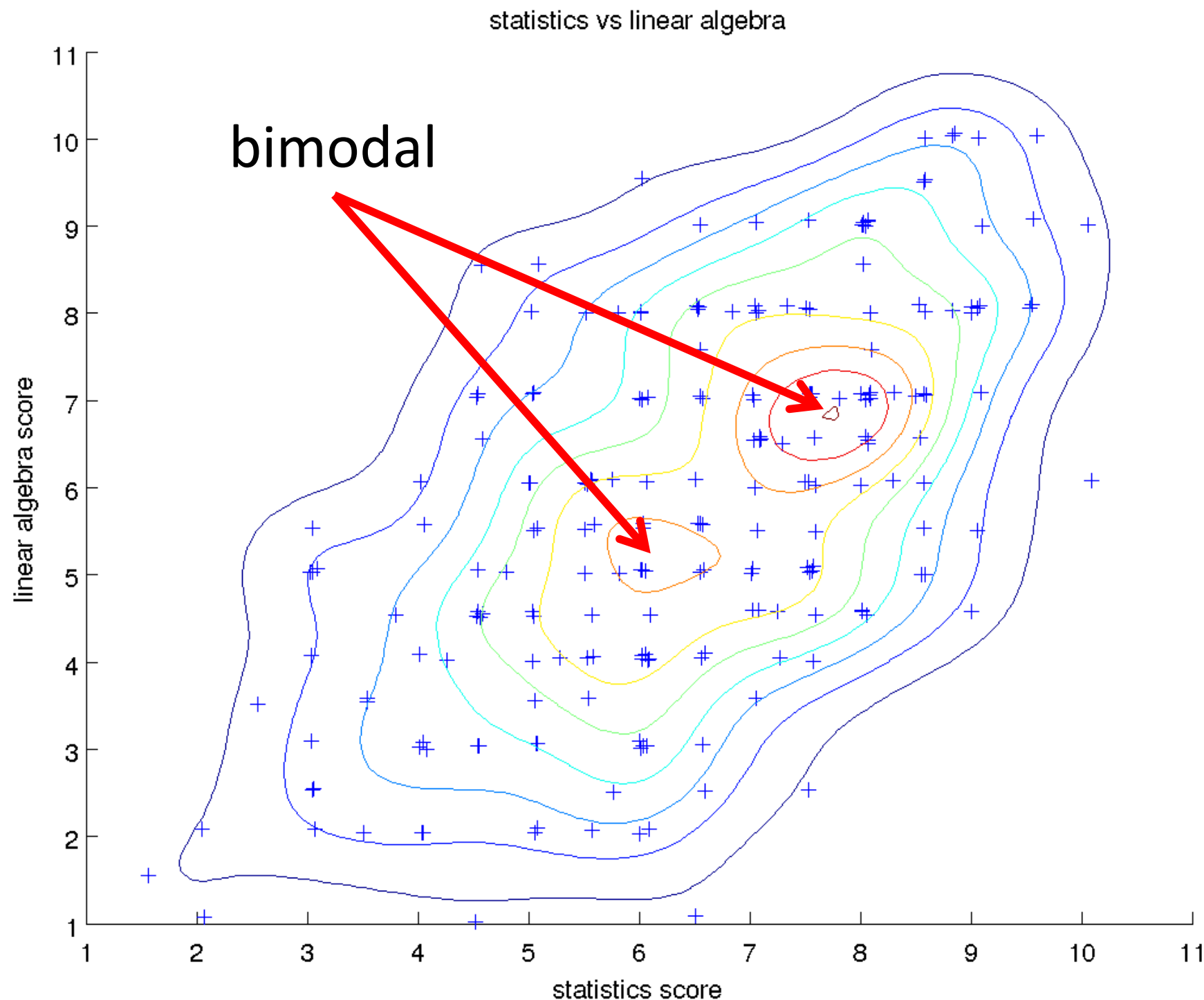
Each set of parent-child relation in graph corresponds to a density (or distribution)

Graphical models for your car

Given the car does not start, and the gas tank 1/10 full, what is the likelihood of leaking?

test scores revisited

- Can we have a more flexible model than Gaussian, but less parameter than KDE?

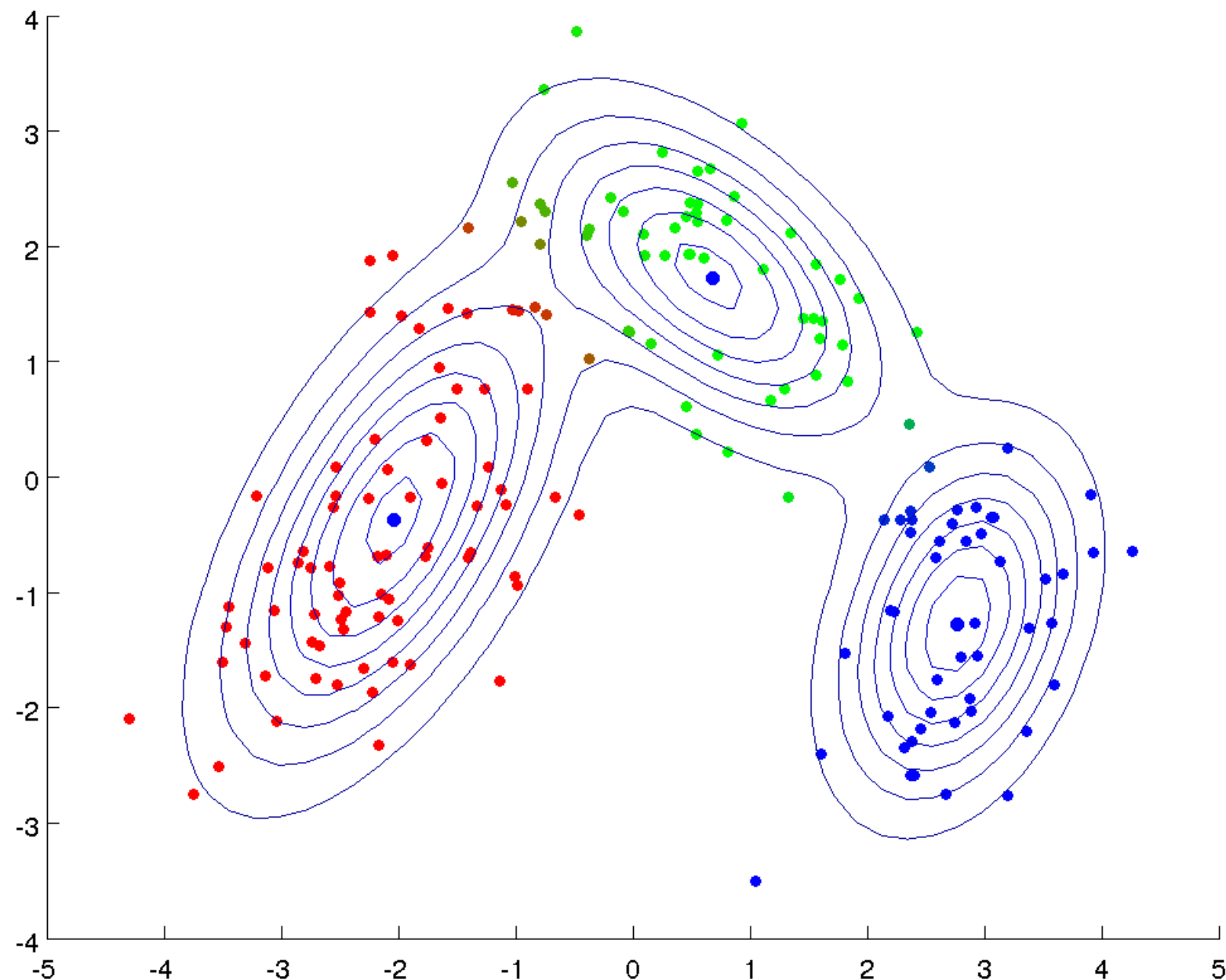


Wine data set <http://archive.ics.uci.edu/ml/datasets/Wine+Quality>

- These data are the results of a chemical analysis of wines grown in the same region in Italy but derived from three different cultivars. Feature include
 - 1) Alcohol
 - 2) Malic acid
 - 3) Ash
 - 4) Alcalinity of ash
 - 5) Magnesium
 - 6) Total phenols
 - 7) Flavanoids
 - 8) Nonflavanoid phenols
 - 9) Proanthocyanins
 - 10) Color intensity
 - 11) Hue
 - 12) OD280/OD315 of diluted wines
 - 13) Proline

Demo: test_wine.m

- First run PCA to reduce the dimension to 2
- Clear cluster structure
- Can we fit 3 Gaussians?



Gaussian mixture model

- A density model $p(X)$ may be multi-modal: model it as a mixture of uni-modal distributions (e.g. Gaussians)

$$\mathcal{N}(X|\mu_k, \Sigma_k) := \frac{1}{|\Sigma|^{1/2}(2\pi)^{d/2}} \exp\left(-\frac{1}{2}(X - \mu)^T \Sigma^{-1}(X - \mu)\right)$$

- Consider a mixture of K Gaussians

- $p(X) = \sum_{k=1}^K \pi_k \mathcal{N}(X|\mu_k, \Sigma_k)$

mixing
proportion

mixture
Component

- Parametric or nonparametric?

- Learn $\pi_k \in (0,1), \mu_k, \Sigma_k$;

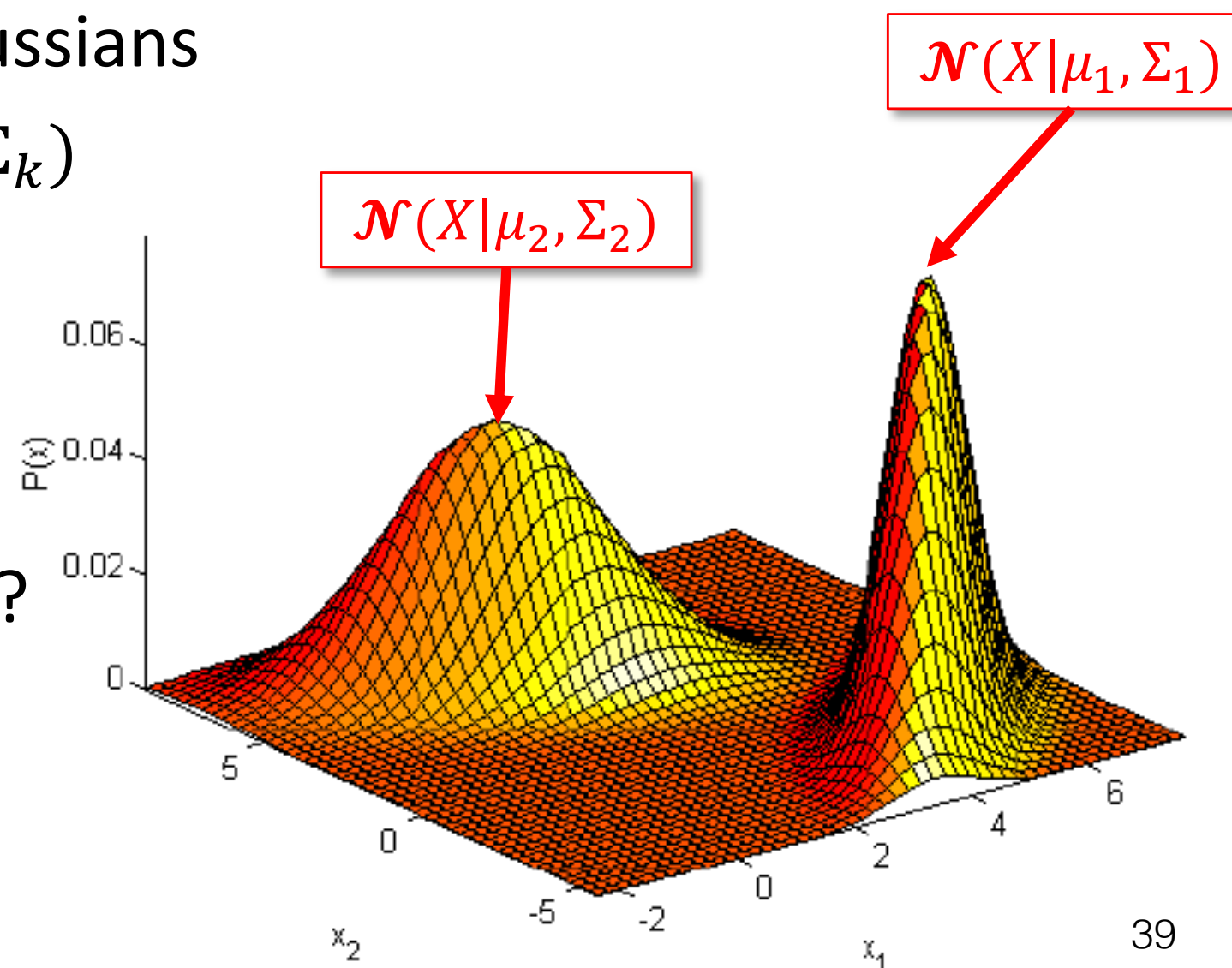


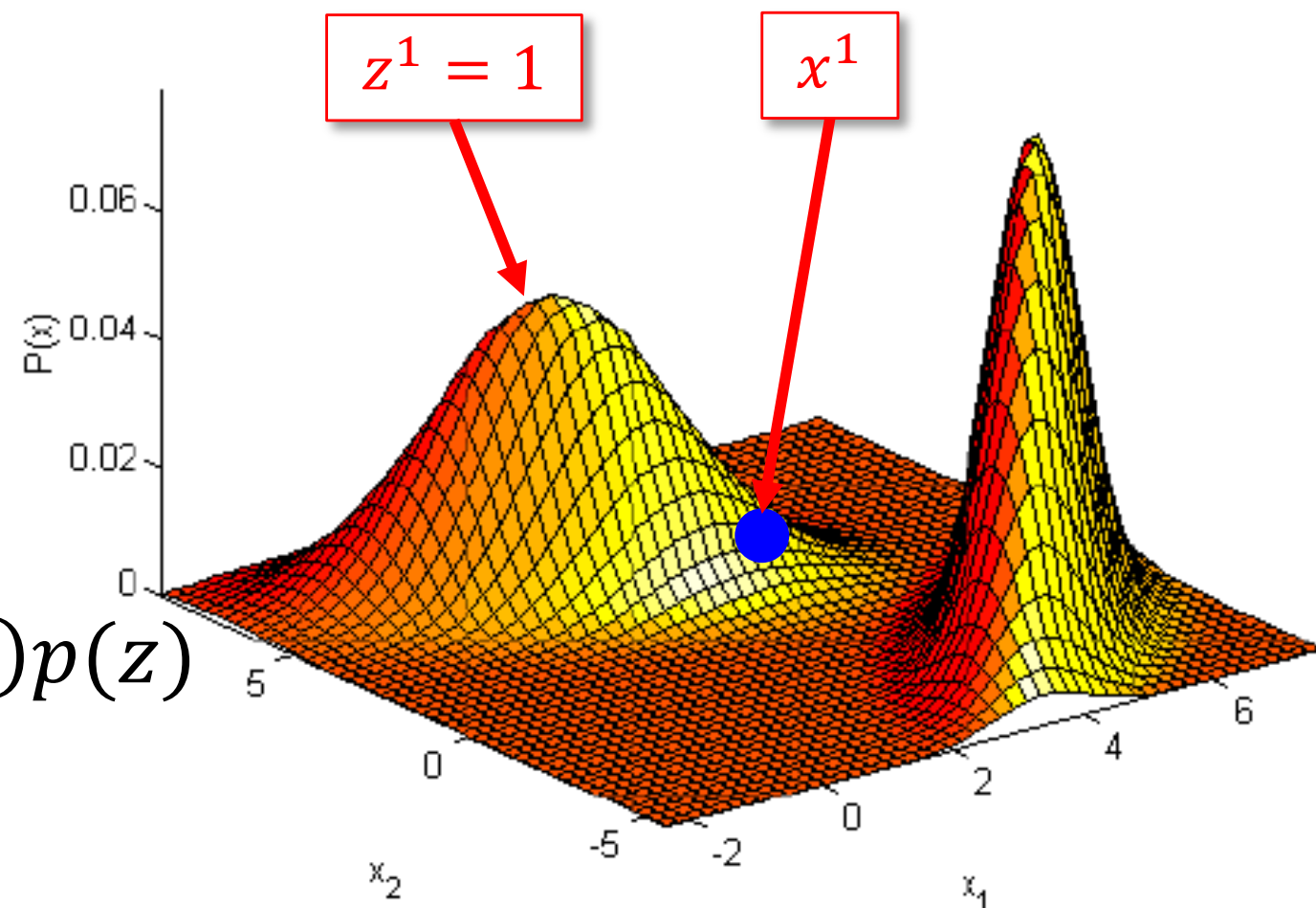
Image a generative process for data points

- For each data point x^i :
 - Randomly choose a mixture component, $z^i = \{1, 2, \dots, K\}$, with probability π_{z^i}
 - Then sample the actual value of x^i from a Gaussian distribution $\mathcal{N}(x | \mu_{z^i}, \Sigma_{z^i})$

- Joint distribution over $p(x, z)$
 $p(x, z) = \pi_z \mathcal{N}(x | \mu_z, \Sigma_z)$

- Marginal distribution $p(x)$

$$p(x) = \sum_{z=1}^K p(x, z) = \sum_{z=1}^K p(x|z)p(z)$$



Learning the parameters

- How to learn?
- Maximum likelihood learning (let $\theta = (\pi_k, \mu_k, \Sigma_k), k = 1 \dots K$)

$$\theta^* = \operatorname{argmax} l(\theta; D) = \log \prod_{i=1}^m p(x^i)$$

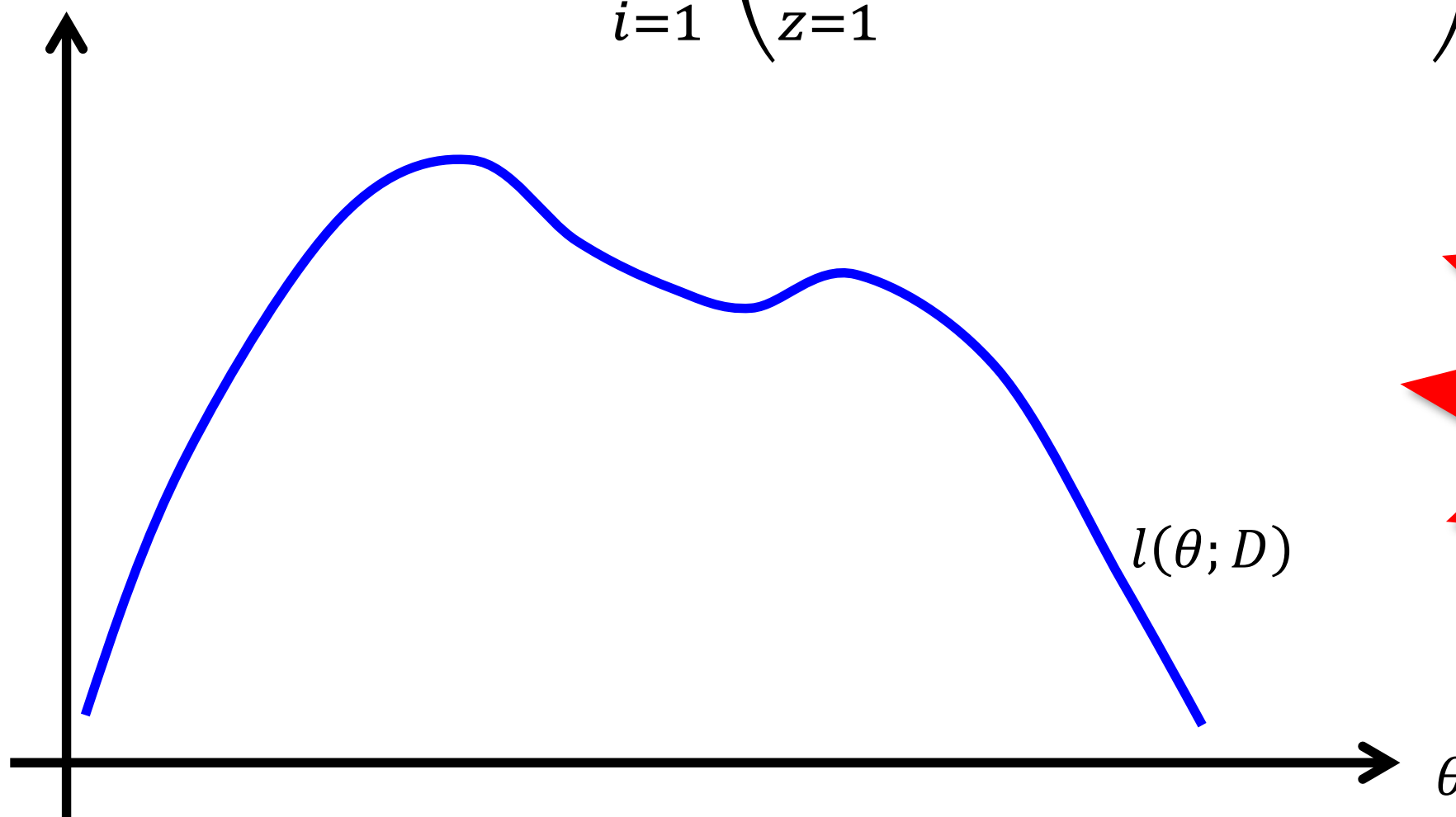
- Use our generative process

$$\begin{aligned} l(\theta; D) &= \log \prod_{i=1}^m \left(\sum_{z^i=1}^K p(x^i, z^i | \theta) \right) \\ &= \log \prod_{i=1}^m \left(\sum_{z^i=1}^K p(x^i | \mu_{z^i}, \Sigma_{z^i}) p(z^i | \pi) \right) \end{aligned}$$

Why is learning hard?

- With latent variables z , likelihood of the data becomes

$$l(\theta; D) = \log \prod_{i=1}^m \left(\sum_{z^i=1}^K p(x^i | \mu_{z^i}, \Sigma_{z^i}) p(z^i | \pi) \right)$$
$$= \log \prod_{i=1}^m \left(\sum_{z=1}^K \pi_{z^i} \mathcal{N}(x | \mu_{z^i}, \Sigma_{z^i}) \right)$$



Nonconvex
Difficult!